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Budget-Driven Multiperiod Hub Location: A Robust Time-Series Approach

Jie Hu,[a,b] Zhi Chen,[c] Shuming Wang[b,d,*]

[a] School of Economics and Management, Beijing Jiaotong University, Beijing 100044, China; [b] School of Economics and Management, University of Chinese Academy of Sciences, Beijing 100190, China; [c] Department of Decisions, Operations and Technology, CUHK Business School, The Chinese University of Hong Kong, Hong Kong; [d] MOE Social Science Laboratory of Digital Economic Forecasts and Policy Simulation, Beijing 100190, China

*Corresponding author

Contact: jiehubuaa@163.com, https://orcid.org/0000-0002-1529-4713 (JH); zhi.chen@cuhk.edu.hk, https://orcid.org/0000-0002-7871-1860 (ZC); wangshuming@ucas.edu.cn, https://orcid.org/0000-0003-0563-8743 (SW)

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Abstract. We study the (un)capacitated multiperiod hub location problem with uncertain periodic demands. With a distributionally robust approach that considers time series, we build a model driven by budgets on periodic costs. In particular, we construct a nested ambiguity set that characterizes uncertain periodic demands via a general multivariate time-series model, and to ensure stable periodic costs, we propose to constrain each expected periodic cost within a budget whereas optimizing the robustness level by maximizing the size of the nested ambiguity set. Statistically, the nested ambiguity set ensures that the model's solution enjoys finite-sample performance guarantees under certain regularity conditions on the underlying VAR(p) or VARMA(p, q) process of the stochastic demand. Operationally, we show that our budget-driven model in the uncapacitated case essentially optimizes a "Sharpe ratio"–type criterion over the worst case among all periods, and we discuss how cost budgets would affect the optimal robustness level. Computationally, the uncapacitated model can be efficiently solved via a bisection search algorithm that solves (in each iteration) a mixed-integer conic program, whereas the capacitated model can be approximated by using decision rules. Finally, numerical experiments demonstrate the attractiveness and competitiveness of our proposed model.

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Keywords: hub location • financial budget • time series • robust optimization • Wasserstein distance

1. Introduction

The hub location problem has been widely studied for designing computer, telecommunication, and transportation hub-and-spoke networks that route commodities between many origin/destination (O/D) pairs (Contreras 2015). Building a hub-and-spoke network for long-run operations (e.g., decades), such as constructing a telecommunication network (Chang and Gavish 1995) and designing the transportation network for biofuel supply chains (Marufuzzaman and Ekşioğlu 2017), is often time-consuming and is usually accomplished in a periodic manner. However, demands between O/D pairs often vary over periods, motivating the multiperiod hub location (MHL) problem that addresses the issues of where and when to locate a set of hubs over the planning horizon (Correia et al. 2018). In this paper, we consider an MHL problem where the demands of customer nodes in each time period are random and are realized sequentially over periods. Before any realization of demands, the decision maker first decides at the beginning of the planning horizon the setup plan for hubs and then, in each period, determines the shipping routes after observing the sequential demand realizations. Two challenges of MHL are addressed in our study: (i) the path-dependent uncertainty of the demand process over periods, and (ii) the risk of cost fluctuation over periods.

1.1. Path-Dependent Uncertainty of the Demand Process

Multiperiod location planning is usually a long-term strategic decision (Qi et al. 2022b). The future demand path over periods is dynamic and uncertain, whereas the underlying demand process is difficult to articulate precisely. It has been well acknowledged that optimization without incorporating uncertainty could significantly hurt the optimal decision in facility location, supply location, freight optimization, as well as other network design problems (Snyder 2006, Unnikrishnan et al. 2009). The majority of the extant literature on location planning under demand uncertainty typically models this long-term planning problem as an adaptive two-stage optimization problem (see, e.g., Wang et al. 2020, Saif and Delage 2021). Existing results mostly deal with only cross-sectional uncertainty and are thus difficult to directly extend to the multiperiod situation that involves the path-dependent stochastic demand process. Because of the more complicated dynamics of uncertainty over time, the MHL problem is more challenging than the two-stage counterpart. In summary, how to develop a statistically and computationally attractive modeling framework that can extrapolate effectively the path-dependent demand uncertainty without sacrificing much tractability remains a crucial research question in the literature on hub location.

1.2. Fluctuation of Periodic Costs

Hub location, as a typical class of projects with a heavy upfront investment, usually requires a relatively stable cash flow over the planning horizon. However, because of the possible highly uncertain demands across periods, the periodic costs could fluctuate significantly. This may result in a heavy financial burden and, consequently, delay or even failure of the project; see, for example, the significant delay of Berlin Brandenburg airport (Thijssen 2022, Wikipedia 2023) and the bankruptcy of the waste recycling company, IUT-Global[1] (The Straits Times 2011, Xiong et al. 2021). Such a temporal risk in the cash flow has been well recognized in many multiperiod planning problems, including investment optimization (Smith 1998), inventory management (Chen et al. 2007), and platform pricing (Ma et al. 2022). When evaluating location decisions, existing literature primarily focuses on the present values of risky cost flow over periods yet overlooks the inherent fluctuation of cost flow that could potentially mislead the decision making. This gap motivates us to mitigate the risk of cost fluctuation over periods—an issue that is not well addressed in the extant hub location literature. To this end, we shall follow the spirit of satisficing (e.g., Brown and Sim 2009, Schwartz et al. 2011, Brown et al. 2012, Chen et al. 2015, Long et al. 2022). In particular, at the beginning of the planning horizon, we prespecify budgets on periodic costs.[2] Such periodic targets are commonly considered in practice, especially for operating a heavyweight project (Mihm 2010, Chen and Tang 2022).

In this paper, we study the (un)capacitated MHL problem with an uncertain demand path over periods using the distributionally robust optimization (DRO) framework. On the one hand, to characterize the *path-dependent uncertainty* on the future periodic demands, we construct a nested ambiguity set that incorporates a general multivariate time-series model (e.g., vector autoregression (VAR) or vector autoregressive moving average (VARMA)) with Wasserstein balls for different periods. On the other hand, to mitigate the *fluctuation of periodic costs*, we propose a budget-driven model that constrains each expected periodic cost within a budget whereas optimizing the robustness of the model by maximizing the size of the nested ambiguity set. Our contributions may be summarized as follows.

1. We model the MHL problem via a Wasserstein DRO framework that incorporates time series. In particular, based on the Wasserstein distance, we construct a nested ambiguity set for a general multivariate time-series model that spans along the planning horizon. Our numerical experiments reveal the value of incorporating time series, which extrapolates the underlying path-dependent stochastic demand process, in the design of an ambiguity set.

2. The proposed nested ambiguity set possesses a desirable property of measure concentration under suitable statistical regularity conditions on the underlying true and the estimated stochastic demand processes as VAR(p) or VARMA(p, q). This then yields attractive finite-sample performance guarantees of our model. To the best of our knowledge, we are the first to investigate the statistical consistency of the Wasserstein time-series ambiguity set under the VARMA(p, q) process (which includes VAR(p) as a special case). Our analysis is nontrivial and might be of independent interest.

3. Our budget-driven model, which ensures a stable cost flow over the planning horizon, has a notable economic interpretation in the uncapacitated situation. By optimizing the robustness level via maximizing the size of the nested ambiguity set, our model essentially maximizes a "Sharpe ratio"–type criterion with respect to the worst case among all periods in the planning horizon. This analytical insight is new to the literature on location under uncertainty.

4. From the computational perspective, our uncapacitated model can be efficiently solved via a bisection search algorithm that solves (in each iteration) a mixed-integer conic program, whereas the capacitated model can be approximated by deploying decision rules.

1.3. Organization and Notation

The remainder of the paper is organized as follows. Section 2 reviews the related literature. In Section 3, we

introduce the budget-driven MHL model and propose the nested ambiguity set. Reformulations and insights are presented in Section 4. We discuss the finite-sample performance guarantees in Section 5. Numerical experiments are presented in Section 6. All proofs are relegated to the Online Appendix.

We denote by $[N] = \{1, 2, \ldots, N\}$ the set of positive running indices from 1 to N, and $[N] = \emptyset$ if $N \leq 0$. Boldface lowercase and uppercase letters denote vectors and matrices, respectively. Special vectors of a proper dimension include $\mathbf{0}$, $\mathbf{1}$, and $\boldsymbol{e}_i$, which represent the vector of all zeros, the vector of all ones, and the ith standard basis. Let $\mathcal{P}(\Xi)$ be the set of all distributions on a subset Ξ of the real space. For a general norm $\|\cdot\|$, its dual norm is $\|\cdot\|_*$. We use $\tilde{\boldsymbol{u}} \sim \mathbb{P}$ to define a random variable $\tilde{\boldsymbol{u}}$ governed by the distribution $\mathbb{P}$, for which $\mathbb{E}_{\mathbb{P}}[\cdot]$ and $\mathbb{P}[\cdot]$ denote the corresponding expectation and probability. Given a logical expression $\mathcal{E}$, $\mathbb{I}[\mathcal{E}]$ is an indicator function that takes a value of one if $\mathcal{E}$ is true, and zero otherwise.

2. Literature Review

We review major streams of related literature: location planning under uncertainty, multistage robust optimization, residual-based DRO approach, and distribution-free stochastic process. We also discuss the research gap between our work and the most relevant existing studies.

2.1. Location Planning Under Uncertainty

The majority of the literature on location planning under uncertainty adopts an adaptive two-stage optimization approach. For instance, Wang et al. (2020) study an adaptive two-stage hub location problem by considering a moment ambiguity set for the uncertain demand and cost. With some assumptions on the dependency between demand and cost, the authors analyze the optimal transportation policies and derive tractable reformulations of the problem. Based on the type 1 Wasserstein ambiguity set, Saif and Delage (2021) consider a two-stage capacitated facility location problem, and they investigate the exact column generation algorithm as well as the approximate decision rule approach.

Some studies consider multistage location planning under demand uncertainty. Correia et al. (2018) develop a stochastic capacitated multiperiod hub location problem given the demand distribution and propose several inequalities for simplifying the problem to enhance the computational scalability. Yu and Shen (2021) formulate a multistage stochastic facility location problem with a demand distribution supported on a finite number of scenarios, and the authors provide tight lower bounds on the gap between the multistage and two-stage models. In addition, the authors explore the optimal priority list of facility locations when facing an uncertain amount of budget (i.e., the maximum number of facilities to be open). Wang et al. (2021a) propose a target-oriented location-transportation problem based on the mean and support information about the demand path over periods and develop a Benders decomposition algorithm to solve the problem. Based on the type-∞ Wasserstein ambiguity set, Wang et al. (2021b) consider a multiperiod facility location problem for humanitarian logistics, where the uncertain demand only occurs at the very beginning of the planning horizon.

Recent interest in demand endogeneity is noteworthy. Basciftci et al. (2021) study a two-stage facility location problem with uncertain endogenous demand and propose a decision-dependent DRO model that admits an equivalent mixed-integer linear programming reformulation. The authors also derive valid inequalities to speed up the solution procedure. For state-dependent demand uncertainty, Liu et al. (2022) formulate an adaptive robust stochastic facility location model that can be solved exactly by a nested Benders decomposition algorithm. The authors further analyze how parameters of the ambiguity set (respectively, location decisions) affect the optimal worst-case expected total cost (respectively, second-stage cost). By extending the moment-based ambiguity set in Delage and Ye (2010) to a multistage and decision-dependent setting, Yu and Shen (2022) investigate multistage distributionally robust mixed-integer programs with endogenous uncertainty and derive mixed-integer semidefinite programming reformulations for stagewise subproblems. Encouraging performances of their reformulations and devised algorithms are showcased and validated in a multistage facility-location problem.

2.2. Multistage Robust Optimization

In the multistage optimization problem under uncertainty, one sequentially makes decisions that are adaptive to (partial) uncertainty realizations over periods (Shapiro 2021). Bertsimas et al. (2011) analyze the performance of the finite adaptability approach for multistage robust optimization problems. Georghiou et al. (2019) propose a robust dual dynamic programming framework for multistage robust optimization problems and offer an approximation scheme that is guaranteed to terminate in a finite number of steps. More recently, Bertsimas et al. (2023) propose a data-driven framework for multistage stochastic linear problems based on historical sample paths, which admits an equivalent interpretation as a DRO model under the type-∞ Wasserstein ambiguity set. The authors also show that the proposed framework can be solved using the decision rule technique. Arora et al. (2022) study a multistage linear DRO model under an uncertain stochastic process that is close to a scenario tree in a nonconvex nested distance and identify several conditions to reformulate the problem as a tractable stochastic program.

It is worth noting that there are some studies emphasizing that the temporal risk plays an important role in the decision making of multiperiod optimization problems (see, e.g., Smith 1998, Mahmoud 2017, Ma et al. 2022). To address the temporal risk in financial decision-making problems, Chen et al. (2015) propose a dynamic model that involves the risky consumption decision subject to a desired target level of consumption in each period over the planning horizon. The aforementioned work of Wang et al. (2021a) also proposes to mitigate the risk of not meeting the summing profit over the whole planning horizon.

2.3. Residual-Based DRO Approach

The residual-based DRO approach has recently attracted a surge of interest in the DRO literature. Roughly speaking, it refers to the DRO approach that first harnesses certain prediction/estimation models and then treats the associated random residuals as the uncertainty. For instance, Dou and Anitescu (2019) propose a (single-stage) DRO model with the data coming from a VAR(1) process, and they capture the ambiguity of the true random noises by a Wasserstein ambiguity set of independent and identically distributed (i.i.d.) residuals. Using the duality argument, the authors reformulate the proposed model as a finite-dimensional convex-concave saddle point problem, and they further discuss the performance guarantees via concentration inequalities. Kannan et al. (2023) introduce a residual-based Wasserstein DRO framework to incorporate the covariate information and analyze finite-sample properties of the model. Perakis et al. (2022) propose an information-partitioning ambiguity set to characterize the random residuals in an additive demand model and reformulate their interested pricing and production problem into a mixed-integer linear program. Qi et al. (2022a) investigate a distributionally robust conditional quantile regression model with a fixed design of the covariates.

2.4. Distribution-Free Stochastic Process

Recent interest in distribution-free stochastic processes is also noteworthy. Sutter et al. (2024) propose a statistically optimal approach that maps the available training data to a feasible decision where the data comes from a time-homogeneous, ergodic finite-state Markov chain. The authors show that if the unknown data-generating distribution belongs to a parametric ambiguity set and the corresponding parameters admit a sufficient statistic satisfying a large deviation principle, the optimal data-driven decision can be obtained from solving a distributionally robust optimization problem constructed from that sufficient statistic and the rate function of its large deviation principle. Li et al. (2021) follow the approach proposed in Sutter et al. (2024) and further show that the aforementioned distributionally robust optimization model is asymptotically consistent. Additionally, to overcome the computational efficiency for the case with large state space, Li et al. (2021) propose a tailored Frank-Wolfe algorithm with convex direction-finding subproblems. Xin and Goldberg (2022) develop a distributionally robust multistage newsvendor model, where the sequence of future demands follows a martingale with known mean and support. With the minimax optimal policy and the corresponding worst-case distribution derived, the authors compare their model with the model of assuming independent demands across periods and identify differences between the two models.

Our work is most closely related to Wang et al. (2021a, b) in the literature on multiperiod location planning, Arora et al. (2022) and Bertsimas et al. (2023) in the literature on multistage robust optimization, Dou and Anitescu (2019), Qi et al. (2022a), and Kannan et al. (2023) in the literature of residual-based DRO approach, and Li et al. (2021), Xin and Goldberg (2022), and Sutter et al. (2024) in the literature of distribution-free stochastic process. We distinguish our work from these related works in detail as follows.

1. In view of multiperiod location planning under uncertainty, Wang et al. (2021a) employ a moment-based ambiguity set to describe the demand path, whereas Wang et al. (2021b), although considering the type-∞ Wasserstein ambiguity set, study only uncertainty whose random components all reveal at once in a single period. In stark contrast, we adopt a time-series approach to capture the demand process, and based on the Wasserstein distance, we use a nested ambiguity set to characterize the uncertain future demand path. In addition, our model possesses finite-sample performance guarantees—a key feature that has not been considered in Wang et al. (2021a) or Wang et al. (2021b).

2. As for the multistage robust optimization subject to uncertainty, we consider time series, take the resulting estimated random errors as uncertainty, and construct a nested ambiguity set based on the type 1 Wasserstein distance. Our approach also establishes finite-sample performance guarantees of the prescribed solutions. Such a model and its related properties for Wasseerstein robust time series have not been analyzed in the works of Arora et al. (2022) and Bertsimas et al. (2023).

3. With respect to the residual-based DRO approach, we first compare against Kannan et al. (2023) and Qi et al. (2022a) that both study finite-sample performance guarantees for their models. Their analysis focuses on a cross-sectional uncertainty with an underlying true value as well as the associated prediction model. In contrast, our finite-sample performance guarantees are developed for our model that considers (multiperiod) multivariate time-series uncertainty with an underlying stochastic process, which has a different yet more

complicated structure and thus involves estimation methods that require more challenging analysis of the measure concentration (see more detailed discussions after Proposition 5). As for the comparison with Dou and Anitescu (2019), we first note that their Wasserstein DRO model under the VAR(1) process is a single-period problem, and the authors set a constraint on the estimated coefficient matrix, whereas we consider a multiperiod problem under the uncertainty of future demands over periods, and we do not constrain the estimators, which results in non-i.i.d. estimated random noises. Second, Dou and Anitescu (2019) assume bounded random noises, whereas we allow unbounded yet light-tailed random noises. Finally, we further analyze the more involved VARMA(p, q) process, and to the best of our knowledge, we are the first to investigate the statistical consistency of the Wasserstein time-series ambiguity set under the VARMA(p, q) process.

4. As for distribution-free stochastic processes, first, both Li et al. (2021) and Sutter et al. (2024) study a single-stage optimization problem involving a finite-state Markov chain. In contrast, we consider an uncountable state space and assume that the demand process follows a time-series process that cannot be recast as a finite-state Markov chain. In addition, we derive the exact (respectively, approximate) solution scheme for our proposed model under the uncapacitated (respectively, capacitated) multiperiod hub location problem, which is distinct from their study purposes. Furthermore, Xin and Goldberg (2022) consider a martingale demand process (for the newsvendor problem) over a mean-support moment ambiguity set capturing that the mean of demand in the current period equals its realization in the last period, whereas we develop a general time-series modeling framework with a nested Wasserstein ambiguity set that characterizes the distributional distance between the true and the estimated demand process via the random noises.

3. Models and Ambiguity Set

We first introduce the mathematical formulation of an uncapacitated multiperiod hub location problem as the base model. We then construct a nested Wasserstein ambiguity set that incorporates a time-series model to characterize the distributional ambiguity of the future demand path over periods. Finally, we formulate the budget-driven distributionally robust counterpart of the problem.

3.1. Base Uncapacitated Model

Consider I potential hubs (indexed by $i \in [I]$), K commodities (indexed by $k \in [K]$), and a planning horizon with T periods (indexed by $t \in [T]$). Let u_{kt} be the demand of commodity k at period t, z_{ijkt} be the shipping decision that is the amount of commodity k shipped via one hub (when $i = j$) or two hubs i and j in period t for each pair of $i, j \in [I]$, and v_{ijkt} be the unit cost of shipping commodity k via hubs i and j at period t. Hubs can be built in different periods with a fixed cost c_{it} of building a facility at site i in period t. Once built, a hub remains open in the subsequent periods.

Let x_{it} be the binary location decision that takes the value of one if and only if hub i is built in period t. For each $t \in [T]$, we denote the location decision and demand in period t by $\boldsymbol{x}_t = (x_{it})_{i\in[I]}$ and $\boldsymbol{u}_t = (u_{kt})_{k\in[K]}$, and we collectively denote the location decisions up to period t by $\mathbf{X}_t = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t) = (x_{is})_{i\in[I], s\le t}$. Given $\mathbf{X}_t$ and $\boldsymbol{u}_t$, the optimal shipping plan in period t with the least shipping cost solves the following problem:

$$\mathrm{Q}_t(\mathbf{X}_t, \boldsymbol{u}_t) = \left[\begin{array}{lll} \text{minimize} & \displaystyle\sum_{k\in[K]}\sum_{i,j\in[I]} v_{ijkt} z_{ijkt} & \\ \text{subject to} & \displaystyle\sum_{i,j\in[I]} z_{ijkt} = u_{kt} & \forall k \in [K] \\ & \displaystyle\sum_{j\in[I]} z_{ijkt} + \sum_{j\in[I]\setminus\{i\}} z_{ijkt} \le \sum_{s\in[t]} x_{is} u_{kt} & \forall k \in [K], i \in [I] \\ & z_{ijkt} \ge 0 & \forall k \in [K], i, j \in [I] \end{array}\right].$$

Here, the first constraint group ensures that each commodity demand should be fully satisfied, whereas the second constraint group imposes that commodities cannot be transported via an unselected hub. An appealing structure of the uncapacitated model is that the shipping cost $\mathrm{Q}_t(\mathbf{X}_t, \boldsymbol{u}_t)$ can be represented as an affine function of demand $\boldsymbol{u}_t$ (see, e.g., Hamacher et al. 2004):

$$\mathrm{Q}_t(\mathbf{X}_t, \boldsymbol{u}_t) = \sum_{k\in[K]} u_{kt} \cdot f_{kt}(\mathbf{X}_t) \quad \forall t \in [T],$$

where, for every $k \in [K]$, $f_{kt}(\mathbf{X}_t)$—which can be interpreted as the shipping cost per unit demand of commodity k in period t given the location decision $\mathbf{X}_t$—is determined by

$$f_{kt}(\mathbf{X}_t) = \left[\begin{array}{lll} \text{minimize} & \displaystyle\sum_{i,j\in[I]} v_{ijkt} y_{ij} & \\ \text{subject to} & \displaystyle\sum_{i,j\in[I]} y_{ij} = 1 & \\ & \displaystyle\sum_{j\in[I]} y_{ij} + \sum_{j\in[I]\setminus\{i\}} y_{ji} \le \sum_{s\in[t]} x_{is} & \forall i \in [I] \\ & y_{ij} \ge 0 & \forall i, j \in [I] \end{array}\right].$$

Here, y_{ij} is the fraction of commodity shipped via the hubs i and j. Such an affine structure plays a key role in facilitating our subsequent analysis and optimization when accounting for randomness in the demand

process. Note that in the current setting, all demands must be satisfied. Nevertheless, the appealing affine structure is still preserved if a penalty on unsatisfied demand is allowed (for instance, any unsatisfied demand of commodity k in period t would incur a unit penalty cost w_{kt}).

The uncapacitated MHL problem, given the deterministic demand, minimizes the total setup cost and shipping cost over T periods, captured by

$$\begin{aligned} \text{minimize} \quad & \sum_{t\in[T]} \tau_t \\ \text{subject to} \quad & \boldsymbol{c}_t^\top \boldsymbol{x}_t + \mathrm{Q}_t(\mathbf{X}_t, \boldsymbol{u}_t) \le \tau_t \quad \forall t \in [T] \\ & \mathbf{X} \in \mathcal{X}, \ \boldsymbol{\tau} \ge \mathbf{0}. \end{aligned} \tag{1}$$

Here, $\boldsymbol{c}_t = (c_{it})_{i\in[I]}$ for all $t \in [T]$, $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_T)$, with τ_t being an epigraphical decision variable that corresponds to the total cost in period t, and $\mathcal{X}$ is the feasible set of the collection of location decisions $\mathbf{X} = \mathbf{X}_T = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T) = (x_{it})_{i\in[I], t\in[T]}$. Throughout our paper, we assume that a site remains open once built, implying $\mathcal{X} = \{\boldsymbol{x} \in \{0,1\}^{I\times T} : \sum_{t\in[T]} x_{it} \le 1 \ \ \forall i \in [I]\}$. Additional requirements on $\mathbf{X}$ (such as the maximum number of hubs to be open) could be imposed in $\mathcal{X}$ straightforwardly.

3.2. Wasserstein Distributionally Robust Time Series

Because periodic demands $\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T$ are revealed sequentially, it is reasonable to view these demands as a time series subject to distributional ambiguity. In particular, we construct an ambiguity set $\mathcal{F}(\theta)$ in this subsection to address the issue of demand ambiguity that the true demand distribution $\mathbb{P}$ of $(\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T)$ is only known to be a member of $\mathcal{F}(\theta)$.

Suppose that a collection of N historical demands $\hat{\boldsymbol{u}}_0, \hat{\boldsymbol{u}}_{-1}, \ldots, \hat{\boldsymbol{u}}_{1-N}$ is available, where, for each $n \in [N]$, $\hat{\boldsymbol{u}}_{1-n} = (u_{1-n,k})_{k\in[K]}$ tracks the demands of commodities $n-1$ periods before now. In addition, we assume that P presamples $\hat{\boldsymbol{u}}_{-N}, \ldots, \hat{\boldsymbol{u}}_{1-N-P}$ are also available, but refer to N as the sample size to avoid notational clutter.[3] Based on the estimation from historical data, the demand forecast model takes the form as follows:

$$\tilde{\boldsymbol{u}}_t = \boldsymbol{a}_t + \mathbf{A}_t^1 \tilde{\boldsymbol{u}}_1 + \mathbf{A}_t^2 \tilde{\boldsymbol{u}}_2 + \cdots + \mathbf{A}_t^{t-1} \tilde{\boldsymbol{u}}_{t-1} + \tilde{\boldsymbol{\xi}}_t \quad \forall t \in [T], \tag{Time Series}$$

where $\boldsymbol{a}_t$ is the intercept, $\mathbf{A}_t^1, \ldots, \mathbf{A}_t^{t-1}$ are matrices of coefficients that capture the relation between $\tilde{\boldsymbol{u}}_t$ and previous demands, and $\tilde{\boldsymbol{\xi}}_t$ is an estimated random noise (ERN). As we will elaborate on in the following examples, Time Series is rich enough to cover many popular time-series models as special cases. In fact, some existing uncertainty modeling approaches for multiperiod settings—such as the growth model in Wang et al. (2016) and the additive demand model in Perakis et al. (2022)—can also be recast via the form of (Time Series).

Example 1 (Vector Autoregression). By multivariate least squares estimation, the p_{th}-order VAR(p) model yields

$$\tilde{\boldsymbol{u}}_t = \boldsymbol{\kappa} + \mathbf{M}_1 \tilde{\boldsymbol{u}}_{t-1} + \cdots + \mathbf{M}_p \tilde{\boldsymbol{u}}_{t-p} + \tilde{\boldsymbol{\xi}}_t,$$

where $(\boldsymbol{\kappa}, \mathbf{M}_1, \ldots, \mathbf{M}_p) = \mathbf{U}\mathbf{Z}^\top (\mathbf{Z}\mathbf{Z}^\top)^{-1}$ with $\mathbf{U} = (\hat{\boldsymbol{u}}_0, \hat{\boldsymbol{u}}_{-1}, \ldots, \hat{\boldsymbol{u}}_{1-N})$ and $\mathbf{Z} = (\mathbf{Z}_{-1}, \ldots, \mathbf{Z}_{-N})$ defined by $\mathbf{Z}_{-n} = (1, \hat{\boldsymbol{u}}_{-n}^\top, \ldots, \hat{\boldsymbol{u}}_{1-n-p}^\top)^\top$ $\forall n \in [N]$. Fitting into (Time Series), we have

$$\tilde{\boldsymbol{u}}_t = \boldsymbol{a}_t + \mathbf{A}_t^1 \tilde{\boldsymbol{u}}_1 + \mathbf{A}_t^2 \tilde{\boldsymbol{u}}_2 + \cdots + \mathbf{A}_t^{t-1} \tilde{\boldsymbol{u}}_{t-1} + \tilde{\boldsymbol{\xi}}_t \quad \forall t \in [T],$$

where $\boldsymbol{a}_t = \boldsymbol{\kappa} + \sum_{s\in[p-t+1]} \mathbf{M}_{t+s-1} \hat{\boldsymbol{u}}_{1-s}$ and $\mathbf{A}_t^s = \mathbb{I}[s \ge t-p] \cdot \mathbf{M}_{t-s}$ for every $s \in [t-1]$.

Example 2 (Vector Autoregressive Moving Average). With two parameters p and q, the VARMA(p, q) model (which covers VAR(p) as a special case) yields

$$\tilde{\boldsymbol{u}}_t = \boldsymbol{\kappa} + \mathbf{M}_1 \tilde{\boldsymbol{u}}_{t-1} + \cdots + \mathbf{M}_p \tilde{\boldsymbol{u}}_{t-p} + \tilde{\boldsymbol{\xi}}_t + \mathbf{L}_1 \tilde{\boldsymbol{\xi}}_{t-1} + \cdots + \mathbf{L}_q \tilde{\boldsymbol{\xi}}_{t-q}.$$

Fitting into (Time Series) (see detailed derivation in Online Appendix F), we have

$$\begin{aligned} \boldsymbol{a}_t = \boldsymbol{\kappa} &+ \sum_{s\in[p-t+1]} \mathbf{M}_{t+s-1} \hat{\boldsymbol{u}}_{1-s} + \sum_{s\in[q-t+1]} \mathbf{L}_{t+s-1} \hat{\tilde{\boldsymbol{\xi}}}_{1-s} \\ &- \sum_{s\in[t-1]} \mathbf{L}_s \boldsymbol{a}_{t-s} \quad \text{and} \\ \mathbf{A}_t^s = & \sum_{\substack{j+i_1+i_2+\cdots=t-s \\ 1\le i_1\le i_2\le\cdots\le q, 1\le j}} \mathbf{L}_{i_1} \mathbf{L}_{i_2} \cdots (\mathbf{M}_j + \mathbf{L}_j) \quad \forall s = 1, \ldots, t. \end{aligned}$$

Here, the number of indices $i_1, i_2, \ldots$ could be zero, and the historical estimated noises $\hat{\tilde{\boldsymbol{\xi}}}_0, \ldots, \hat{\tilde{\boldsymbol{\xi}}}_{1-q}$ are calculated from historical demands.

Equipped with (Time Series), we next capture the distributional ambiguity of ERNs over periods. In particular, we assume that for each $t \in [T]$, the ERN $\tilde{\boldsymbol{\xi}}_t$ is governed by some distribution $\mathbb{Q}_t$, which is ambiguous because of difficulties in the estimation procedure but is known to be close to some reference $\hat{\mathbb{Q}}_t$ that is consistent with the available information. In particular, we measure the closeness between $\mathbb{Q}_t$ and $\hat{\mathbb{Q}}_t$ by the following (type 1) Wasserstein distance (see, e.g., Mohajerin Esfahani and Kuhn 2018, Gao and Kleywegt 2023):

$$\begin{aligned} d(\mathbb{Q}_t, \hat{\mathbb{Q}}_t) = \inf_{\Pi} \{ & \mathbb{E}_\Pi[\|\tilde{\boldsymbol{\xi}}_t - \tilde{\boldsymbol{\xi}}_t'\|] : \Pi \in \mathcal{P}(\mathbb{R}^K \times \mathbb{R}^K) \\ & \text{with marginals } \mathbb{Q}_t \text{ and } \hat{\mathbb{Q}}_t, \text{ respectively} \}, \end{aligned}$$

where $\tilde{\boldsymbol{\xi}}_t \sim \mathbb{Q}_t$, $\tilde{\boldsymbol{\xi}}_t' \sim \hat{\mathbb{Q}}_t$, $\mathcal{P}(\mathbb{R}^K \times \mathbb{R}^K)$ represents the set of all joint distributions supported on $\mathbb{R}^K \times \mathbb{R}^K$, and $\|\cdot\|$ represents a general norm (e.g., the L_ℓ-norm $\|\cdot\|_\ell$ and the Mahalanobis norm). The distributional ambiguity of $\tilde{\boldsymbol{\xi}}_t$ is then captured by the fact that $\mathbb{Q}_t$ is only known to reside in a θ-neighborhood around $\hat{\mathbb{Q}}_t$ with respect to

the Wasserstein distance, that is, $\mathbb{Q}_t \in \mathcal{B}_t(\theta)$ such that

$$\mathcal{B}_t(\theta) = \{\mathbb{Q} \in \mathcal{P}(\mathbb{R}^K) \mid d(\mathbb{Q}, \hat{\mathbb{Q}}_t) \le \theta\}.$$

Here, $\mathcal{B}_t(0) = \{\hat{\mathbb{Q}}_t\}$ because $d(\mathbb{Q}_t, \hat{\mathbb{Q}}_t) = 0$ only if $\mathbb{Q}_t = \hat{\mathbb{Q}}_t$.

We next model, in a nested fashion, the distributional ambiguity of demands $\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T$. Recall from (Time Series) that for $t = 1$, the distribution $\mathbb{P}_1$ of $\tilde{\boldsymbol{u}}_1$ is determined by the constant $\boldsymbol{a}_1$ and the ERN $\tilde{\boldsymbol{\xi}}_1$. Thus, the ambiguity set of $\tilde{\boldsymbol{u}}_1$ can be captured by

$$\mathcal{F}_1(\theta) = \left\{ \mathbb{P}_1 \in \mathcal{P}(\mathbb{R}^K_+) \;\middle|\; \begin{array}{l} \tilde{\boldsymbol{u}}_1 \sim \mathbb{P}_1, \tilde{\boldsymbol{\xi}}_1 \sim \mathbb{Q}_1 \\ \tilde{\boldsymbol{u}}_1 = \boldsymbol{a}_1 + \tilde{\boldsymbol{\xi}}_1 \\ \mathbb{Q}_1 \in \mathcal{B}_1(\theta) \end{array} \right\}.$$

Then, for $t = 2, \ldots, T$, the ambiguity set of $\tilde{\boldsymbol{u}}_t$ conditioning on the realization $\mathbf{U}_{t-1} = (\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{t-1})$ of $(\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_{t-1})$, denoted by $\mathcal{F}_{t|\mathbf{U}_{t-1}}$, is

$$\mathcal{F}_{t|\mathbf{U}_{t-1}}(\theta) = \left\{ \mathbb{P}_{t|\mathbf{U}_{t-1}} \in \mathcal{P}(\mathbb{R}^K_+) \;\middle|\; \begin{array}{l} \tilde{\boldsymbol{u}}_t \sim \mathbb{P}_{t|\mathbf{U}_{t-1}}, \tilde{\boldsymbol{\xi}}_t \sim \mathbb{Q}_t \\ \tilde{\boldsymbol{u}}_t = \boldsymbol{a}_t + \mathbf{A}^1_t \boldsymbol{u}_1 + \cdots + \mathbf{A}^{t-1}_t \boldsymbol{u}_{t-1} + \tilde{\boldsymbol{\xi}}_t \\ \mathbb{Q}_t \in \mathcal{B}_t(\theta) \end{array} \right\},$$

where $\mathbb{P}_{t|\mathbf{U}_{t-1}}$ is the conditional distribution of $\tilde{\boldsymbol{u}}_t$ given $\mathbf{U}_{t-1}$.

Finally, we formulate the ambiguity set for the true distribution $\mathbb{P}$ of the future demand path $(\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T)$ as follows:

$$\mathcal{F}(\theta) = \left\{ \mathbb{P} \in \mathcal{P}(\mathbb{R}^{K\times T}_+) \;\middle|\; \begin{array}{ll} (\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T) \sim \mathbb{P} & \\ \tilde{\boldsymbol{u}}_1 \sim \mathbb{P}_1, \tilde{\boldsymbol{u}}_{t|\mathbf{U}_{t-1}} \sim \mathbb{P}_{t|\mathbf{U}_{t-1}} & \forall \mathbf{U}_{t-1} \in \mathbb{R}^K_+, \\ & \forall t = 2, \ldots, T \\ \mathbb{P}_1 \in \mathcal{F}_1(\theta), \mathbb{P}_{t|\mathbf{U}_{t-1}} \in \mathcal{F}_{t|\mathbf{U}_{t-1}}(\theta) & \forall \mathbf{U}_{t-1} \in \mathbb{R}^K_+, \\ & \forall t = 2, \ldots, T \end{array} \right\}. \tag{Nested}$$

Finally, we present some candidates for the reference distribution of the ERNs.

Example 3 (Gaussian Distribution). In the VARMA(p, q) process, it is usually assumed that the ERN follows a Gaussian distribution of zero mean and an estimated covariance matrix Σ that can be determined by historical ERNs, for instance, via $\boldsymbol{\Sigma} = 1/N \sum_{n\in[N]} \hat{\boldsymbol{\xi}}_{1-n} \hat{\boldsymbol{\xi}}^\top_{1-n}$ (see also section 3.2 of Lütkepohl 2005). Therefore, we can set $\hat{\mathbb{Q}}_t = \mathcal{N}(\mathbf{0}, \Sigma)$ for every $t \in [T]$.

Example 4 (Empirical Distribution). Using past demand observations, the historical ERNs $\hat{\boldsymbol{\xi}}_{1-n}, n \in [N]$ can be determined, for instance, via $\hat{\boldsymbol{\xi}}_{1-n} = \hat{\boldsymbol{u}}_{1-n} - \boldsymbol{\kappa} - \mathbf{M}_1 \hat{\boldsymbol{u}}_{-n}$ for a VAR(1) process. We can then form, for every $t \in [T]$, an empirical distribution $\hat{\mathbb{Q}}_t = 1/N \sum_{n\in[N]} \delta_{\hat{\boldsymbol{\xi}}_{1-n}}$ as the reference distribution of $\mathcal{B}_t(\theta)$, where $\delta_{\hat{\boldsymbol{\xi}}_{1-n}}$ is a Dirac measure that places a unit mass at $\hat{\boldsymbol{\xi}}_{1-n}$.

Example 5 (General Distribution). We can also consider a general distribution as the reference distribution $\hat{\mathbb{Q}}_t$ of $\mathcal{B}_t(\theta)$ for every $t \in [T]$. In this case, the uncapacitated ($\mathcal{BDO}$) model admits a tractable reformulation as long as the mean of $\hat{\mathbb{Q}}_t$ is known (Section 4). As for the capacitated model in Online Appendix E, we may construct, for the sake of computational tractability, an approximate discrete reference distribution with samples being generated from $\hat{\mathbb{Q}}_t$.

Remark 1. Some existing studies (e.g., Dou and Anitescu 2019) utilize the ambiguity set equipped with the time-series model to characterize the demand randomness and differentiate the parameter uncertainty from the distributional ambiguity of i.i.d. noises. Our approach captures instead the distributional ambiguity of non-i.i.d. ERNs. Dou and Anitescu (2019) focus on single-period decision making and consider a demand process with *bounded* noises, whereas we consider a more general demand process with *unbounded* noises under the light tail assumption, requiring more complicated analysis (to be introduced in Section 5) to establish our ambiguity set's measure concentration. Applying both approaches to the uncapacitated single-period hub location problem yields tractable reformulations of similar structures. Yet the approach of Dou and Anitescu (2019) takes a longer time to validate, as it involves more hyperparameters. More importantly, we would like to emphasize that extending to multiperiod problems, the approach proposed by Dou and Anitescu (2019) would induce the issue of bilinearity. We refer to Online Appendix C.4 for more detailed discussions.

3.3. Budget-Driven Uncapacitated Model

For sustainable operations management, it is of practical importance to control the cost level of each period. This motivates us to prespecify for each $t \in [T]$, τ_t as the cost budget in period t in Problem (1). Given the periodic budgets $\boldsymbol{\tau}$, for a location decision $\mathbf{X}$, we are interested in its capability to ensure that its related periodic expected costs do not violate the corresponding budgets. In particular, we note that the parameter $\theta \ge 0$ of our proposed (Nested) ambiguity set $\mathcal{F}(\theta)$ indicates its size in the sense that $\mathcal{F}(\theta_1) \subseteq \mathcal{F}(\theta_2)$ for any $\theta_2 \ge \theta_1 \ge 0$. Hence, we identify the maximal size of the ambiguity set with respect to which the worst-case periodic expected costs of $\mathbf{X}$ meet the budgets. The maximal size, denoted by $\theta^\star(\mathbf{X})$, can be captured by

$$\theta^\star(\mathbf{X}) = \left[\begin{array}{ll} \text{maximize} & \theta \\ \text{subject to} & Z_t(\mathbf{X}, \theta) = \sup\limits_{\mathbb{P}\in\mathcal{F}(\theta)} \mathbb{E}_{\mathbb{P}}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + Q_t(\mathbf{X}_t, \tilde{\boldsymbol{u}}_t)] \\ & \qquad \le \tau_t \quad \forall t \in [T] \\ & \theta \ge 0 \end{array} \right]. \tag{2}$$

Here, $Z_t(\mathbf{X},\theta)$ is the worst-case expected periodic cost under distributional ambiguity sized by θ. Therefore, $\theta^\star(\mathbf{X})$ stands for the robustness level of the location decision $\mathbf{X}$, and a larger value of $\theta^\star(\mathbf{X})$ indicates a more robust capability of $\mathbf{X}$ in meeting the periodic budgets $\boldsymbol{\tau}$ whereas hedging against distributional ambiguity. Whereas there are some existing works (e.g., Yu and Shen 2021) that treat facility location decisions as recourse decisions, in our model, the plan of where and when to locate a set of hubs is decided at the beginning of the planning horizon. On the one hand, such a setting of static location decisions simplifies the model and thus makes our subsequent budget-driven models tractable. On the other hand, in some situations, constructing a hub-and-spoke network as a heavyweight project requires a specific plan over the whole horizon at the beginning of the project.

Equipped with $\theta^\star(\mathbf{X})$, given the periodic budgets $\boldsymbol{\tau}$, we then seek the optimal (i.e., largest) robustness level, denoted by $\theta^\star$, via the following *budget-driven* optimization model:[4]

$$\begin{array}{ll} \text{maximize} & \theta^\star(\mathbf{X}) \\ \text{subject to} & \mathbf{X}\in\mathcal{X}. \end{array} \qquad (\mathcal{BDO})$$

Determining the periodic budgets—the key inputs of our ($\mathcal{BDO}$) model—involves the following considerations. In practice, several potential budget proposals at the project level are typically provided by project managers together with work-center managers.[5] Such proposals are usually based on the prediction of some key operational information (e.g., demand and cash flow), possibly adjusted by the managers' preferences and judgments (Chen and Tang 2022). For instance, one can first forecast periodic demands and then estimate the corresponding periodic budgets under the demand forecast. Moreover, to improve the quality of periodic budgets, we can apply cross-validation (see, e.g., Long et al. 2022) to tune them over the set of budget proposals; see more details in our experiments of Section 6.1. Periodic budgets may be considered uncertain in some situations (see, e.g., Tsetlin and Winkler 2007, Keisler and Bordley 2015). Our ($\mathcal{BDO}$) model can also incorporate uncertain periodic budgets in several manners; see Online Appendix C.3 for more details. Note that some existing works in facility locations interpret the budget as the maximum number of facilities to be open (e.g., Basciftci et al. 2021, Yu and Shen 2021). Our ($\mathcal{BDO}$) model sets, for each period, the budget for the total cost of setup and shipping, thus differentiating from this stream.

Remark 2 (Traditional DRO Counterpart). In contrast to our proposed ($\mathcal{BDO}$) model, the traditional distributionally robust optimization approach equipped with the (Nested) ambiguity set minimizes the worst-case expected total cost as follows:

$$\begin{array}{ll} \text{minimize} & \tau \\ \text{subject to} & \sup\limits_{\mathbb{P}\in\mathcal{F}(\theta)} \mathbb{E}_{\mathbb{P}}\left[\sum\limits_{t\in[T]}(\boldsymbol{c}_t^\top \boldsymbol{x}_t + \mathrm{Q}_t(\mathbf{X}_t,\tilde{\boldsymbol{u}}_t))\right] \le \tau \\ & \mathbf{X}\in\mathcal{X},\ \tau\in\mathbb{R}. \end{array} \qquad (\mathcal{DRO})$$

From the computational perspective, ($\mathcal{BDO}$) shares similar computational complexity with ($\mathcal{DRO}$); see Online Appendix C.1 for the reformulation of ($\mathcal{DRO}$). From the modeling perspective, ($\mathcal{BDO}$) emphasizes and mitigates the fluctuation of periodic costs. Meanwhile, because the periodic budgets imposed in the ($\mathcal{BDO}$) model induce more constraints, as a trade-off, it may lead to a higher total cost for the more stable periodic cost flows. These are also justified by our experiments in Section 6.1.

To conclude this section, we point out that our ($\mathcal{BDO}$) model can also be extended to the capacitated case. Compared with the uncapacitated case, the budget-driven capacitated MHL is more difficult to solve. To achieve a computationally viable solution, we resort to decision rule approximation (see also Chen and Zhang 2009, de Ruiter 2018, Bertsimas et al. 2019, Chen et al. 2020). We refer to Online Appendix E for more details about budget-driven capacitated MHL.

4. Reformulation and Sensitivity Analysis

In this section, we present an insightful interpretation of the ($\mathcal{BDO}$) model and derive its tractable reformulation. We also analyze the impact of the periodic budgets on the optimal robustness level.

4.1. Insights and Tractable Reformulation

Exploring the ambiguity set $\mathcal{F}(\theta)$ in (Nested), we can reexpress the worst-case expected periodic costs in the $\mathcal{BDO}$ model with a nested form as well.

Proposition 1. *Given $\mathcal{F}(\theta)$ in* (Nested), *for all $t\in[T]$, we have*

$$\mathrm{Z}_t(\mathbf{X},\theta) = \sup_{\mathbb{P}_1\in\mathcal{F}_1(\theta)} \mathbb{E}_{\mathbb{P}_1}\left[\sup_{\mathbb{P}_2\in\mathcal{F}_{2|\mathbf{U}_1}(\theta)} \mathbb{E}_{\mathbb{P}_2}\left[\cdots \sup_{\mathbb{P}_t\in\mathcal{F}_{t|\mathbf{U}_{t-1}}(\theta)} \mathbb{E}_{\mathbb{P}_t}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \mathrm{Q}_t(\mathbf{X}_t,\tilde{\boldsymbol{u}}_t)\,|\,\mathbf{U}_{t-1}]\cdots|\,\mathbf{U}_1\right]\right]. \qquad (3)$$

The Nested Presentation (3) involves the propagation of demand uncertainty over periods, which appears to lead to the "curse of dimensionality" that is common in multistage optimization problems (Shapiro et al. 2021). Fortunately, owing to the affine structure of the

objective function of the uncapacitated MHL problem (i.e., $Q_t(\mathbf{X}_t, \boldsymbol{u}_t) = \sum_{k\in[K]} u_{kt} f_{kt}(\mathbf{X}_t)$ for all $t \in [T]$) and the structure of the proposed (Nested) ambiguity set, we are able to rearrange terms in (3) based on each $\tilde{\boldsymbol{\xi}}_t$ and obtain a regular DRO reformulation of $Z_t(\mathbf{X}, \theta)$.

Proposition 2. *Given $\mathcal{F}(\theta)$ in (Nested), for all $t \in [T]$, we have*

$$Z_t(\mathbf{X},\theta) = \sup_{\mathbb{Q}_{[t]}\in\mathcal{B}_{[t]}(\theta)} \mathbb{E}_{\mathbb{Q}_{[t]}}\left[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \boldsymbol{\eta}_t^\top \boldsymbol{f}_t(\mathbf{X}_t) + \sum_{s\in[t]} [\mathbf{B}_t^s \tilde{\boldsymbol{\xi}}_s]^\top \boldsymbol{f}_t(\mathbf{X}_t)\right]. \tag{4}$$

Here, $\boldsymbol{f}_t(\cdot) = (f_{kt}(\cdot))_{k\in[K]}$, $\mathbb{Q}_{[t]} = \otimes_{s\in[t]}\mathbb{Q}_s$, $\mathcal{B}_{[t]}(\theta) = \otimes_{s\in[t]}\mathcal{B}_s(\theta)$, and the deterministic vector $\boldsymbol{\eta}_t$ and matrices $\mathbf{B}_t^s$ from an estimated (Time Series), given as

$$\boldsymbol{\eta}_t = \boldsymbol{a}_t + \sum_{s\in[t-1]} \mathbf{A}_t^s \boldsymbol{\eta}_s \quad \textit{and}$$

$$\mathbf{B}_t^s = \sum_{s<i_1<i_2<\cdots<i_l<t-1} \mathbf{A}_{i_1}^s \mathbf{A}_{i_2}^{i_1} \cdots \mathbf{A}_t^{i_l} \quad \forall s = 1,\ldots,t-1,$$

are assumed to be nonnegative, and $\mathbf{B}_t^t$ is a K-dimensional identity matrix.

For each $t \in [T]$, the term in the Worst-Case Expectation (4) is affine in $\tilde{\boldsymbol{\xi}}_t$. This enables us to employ the regularization result (theorem 6.3 of Mohajerin Esfahani and Kuhn 2018 or corollary 2 of Gao et al. 2022) to derive an insightful closed-form representation that we formalize as follows.

Theorem 1. *The ($\mathcal{BDO}$) model is equivalent to*

$$\theta^\star = \left[\begin{array}{ll} \text{maximize} & \min_{t\in[T]} \gamma_t(\mathbf{X}) = \dfrac{(\tau_t - \mathbb{E}_{\hat{\mathbb{Q}}_{[t]}}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \tilde{\boldsymbol{u}}_t^\top \boldsymbol{f}_t(\mathbf{X}_t)])^+}{\nabla_\theta Z_t(\mathbf{X},\theta)} \\ \text{subject to} & \mathbf{X}\in\mathcal{X} \end{array}\right], \tag{5}$$

where $\hat{\mathbb{Q}}_{[t]} = \otimes_{s\in[t]}\hat{\mathbb{Q}}_s$, $\nabla_\theta Z_t(\mathbf{X},\theta) = \sum_{s\in[t]} \|[\mathbf{B}_t^s]^\top \boldsymbol{f}_t(\mathbf{X}_t)\|_*$ is the gradient of $Z_t(\mathbf{X},\theta)$ at θ.

Implications of Theorem 1 are twofold. First, Reformulation (5) reveals that for each period t, the corresponding component $\gamma_t(\mathbf{X})$ in the objective function of ($\mathcal{BDO}$) is a Sharpe ratio–type criterion on "the expected cost below budget under the reference distribution $\hat{\mathbb{Q}}_{[t]}$" over "the sensitivity of the worst-case expected cost $Z_t(\mathbf{X},\theta)$ to the size θ of distributional ambiguity of ERNs." The optimal robustness level $\theta^\star$ then optimizes such a criterion over the worst case among all T periods. Such a Sharpe ratio–type criterion, in fact, generalizes the classical Sharpe ratio (Sharpe 1994).

Corollary 1. *Suppose that for all $t \in [T]$, $\hat{\mathbb{Q}}_t = \mathcal{N}(\mathbf{0}, \Sigma)$ and $\|\cdot\|$ in $\mathcal{B}_t(\theta)$ is the Mahalanobis norm based on the estimated covariance matrix Σ such that $\|\boldsymbol{b}\| = \|\Sigma^{-1/2}\boldsymbol{b}\|_2$. Then, for all $t \in [T]$,*

$$\gamma_t(\mathbf{X}) = \frac{(\tau_t - \mathbb{E}_{\hat{\mathbb{Q}}_{[t]}}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \tilde{\boldsymbol{u}}_t^\top \boldsymbol{f}_t(\mathbf{X}_t)])^+}{\sum_{s\in[t]} \sqrt{\mathbb{V}\mathrm{ar}_{\hat{\mathbb{Q}}_s}[[\mathbf{B}_t^s \tilde{\boldsymbol{\xi}}_s]^\top \boldsymbol{f}_t(\mathbf{X}_t)]}},$$

where $\mathbb{V}\mathrm{ar}_{\hat{\mathbb{Q}}_t}[\cdot]$ is the variance under distribution $\hat{\mathbb{Q}}_t$. In the single-period case (i.e., $T = 1$),

$$\gamma_1(\mathbf{X}) = \frac{(\tau_1 - \mathbb{E}_{\hat{\mathbb{Q}}_1}[\boldsymbol{c}_1^\top \boldsymbol{x}_1 + \tilde{\boldsymbol{u}}_1^\top \boldsymbol{f}_1(\boldsymbol{x}_1)])^+}{\sqrt{\mathbb{V}\mathrm{ar}_{\hat{\mathbb{Q}}_1}[\boldsymbol{c}_1^\top \boldsymbol{x}_1 + \tilde{\boldsymbol{u}}_1^\top \boldsymbol{f}_1(\boldsymbol{x}_1)]}}.$$

Second, Reformulation (5) is also computationally appealing. Observe that the robustness level $\theta^\star(\mathbf{X})$ of a location decision, $\mathbf{X}$ can be evaluated through

$$\theta^\star(\mathbf{X}) = \min_{t\in[T]} \gamma_t(\mathbf{X}) = \min_{t\in[T]} \left\{\frac{(\tau_t - \mathbb{E}_{\hat{\mathbb{Q}}_{[t]}}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \tilde{\boldsymbol{u}}_t^\top \boldsymbol{f}_t(\mathbf{X}_t)])^+}{\sum_{s\in[t]} \|[\mathbf{B}_t^s]^\top \boldsymbol{f}_t(\mathbf{X}_t)\|_*}\right\}.$$

Optimizing the robustness level over $\mathbf{X} \in \mathcal{X}$ can then be achieved by a bisection search wherein each iteration examines the feasibility of a mixed-integer conic constraint system. This is formalized in the subsequent theorem and algorithm.

Theorem 2. *Given $\mathcal{F}(\theta)$ in (Nested), the feasible set of ($\mathcal{BDO}$) is equivalent to the following mixed-integer conic constraint system:*

$$\left\{\begin{array}{ll} \boldsymbol{c}_t^\top \boldsymbol{x}_t + \boldsymbol{\eta}_t^\top \boldsymbol{h}_t + \sum_{s\in[t]} \boldsymbol{h}_t^\top \mathbf{B}_t^s \bar{\boldsymbol{\xi}}_s + \theta\lambda_t^s \le \tau_t & \forall t\in[T] \\ \|[\mathbf{B}_t^s]^\top \boldsymbol{h}_t\|_* \le \lambda_t^s & \forall t\in[T], s\in[t] \\ \sum_{i,j\in[I]} v_{ijkt} y_{ijkt} \le h_{kt} & \forall t\in[T], k\in[K] \\ \sum_{i,j\in[I]} y_{ijkt} = 1 & \forall t\in[T], k\in[K] \\ \sum_{j\in[I]} y_{ijkt} + \sum_{j\in[I]\setminus\{i\}} y_{jikt} \le \sum_{s\in[t]} x_{is} & \forall t\in[T], k\in[K], i\in[I] \\ \lambda_t^s \in \mathbb{R}_+, y_{ijkt} \in \mathbb{R}_+, \boldsymbol{h}_t \in \mathbb{R}_+^K & \forall t\in[T], k\in[K], \\ & \quad i,j\in[I], s\in[t] \\ \mathbf{X}\in\mathcal{X}, \theta\ge 0, & \end{array}\right. \tag{6}$$

where, for each $t \in [T]$, $\bar{\boldsymbol{\xi}}_t = \mathbb{E}_{\hat{\mathbb{Q}}_t}[\tilde{\boldsymbol{\xi}}_t]$.

The pseudocode of the bisection search algorithm is summarized as follows.

Algorithm 1 (Bisection Search Algorithm of the Optimal Robustness Level $\theta^\star$ of ($\mathcal{BDO}$))

Input: $\underline{\theta} = 0$, some suitably large $\overline{\theta}$, and error tolerance ε.

Repeat

1. $\theta = (\underline{\theta} + \overline{\theta})/2$.

2. Check the feasibility of the mixed-integer conic constraint system (6).
3. If (6) is feasible, $\underline{\theta} = \theta$; else, $\overline{\theta} = \theta$.

Until $\overline{\theta} - \underline{\theta} \le \varepsilon$.
Output: $\theta^\star = (\underline{\theta} + \overline{\theta})/2$.

Remark 3. Feasibility of the Constraint System (6) can be checked by solving a mixed-integer linear program if $\|\cdot\|$ is taken as the L_1-norm or L_∞-norm and a mixed-integer second-order cone program if $\|\cdot\|$ is taken as the L_ℓ-norm with some finite rational number $\ell > 1$ (see section 3.3.1 of Ben-Tal and Nemirovski 2001).

4.2. Impact of Periodic Budgets on the Optimal Robustness Level

The periodic budgets could significantly affect the optimal robustness level, which is of practical relevance for applying the proposed ($\mathcal{BDO}$) model. In this section, we investigate the sensitivity of the optimal robustness level $\theta^\star$ to the periodic budgets $\boldsymbol{\tau}$. Our analysis is closely related to the following subset of periods that correspond to the worst-case robustness level:

$$\mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X}) = \arg\min_{t\in[T]} \frac{(\tau_t - \mathbb{E}_{\hat{\mathbb{Q}}_{[t]}}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \tilde{\boldsymbol{u}}_t^\top \boldsymbol{f}_t(\mathbf{X}_t)])^+}{\sum_{s\in[t]} \|[\mathbf{B}_t^s]^\top \boldsymbol{f}_t(\mathbf{X}_t)\|_*}.$$

In particular, our argument involves two cases: (i) $|\mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X})| = 1$ and (ii) $|\mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X})| \ge 2$, indicating whether there is a unique period that corresponds to the worst-case robustness level.

Theorem 3. *Suppose that* ($\mathcal{BDO}$) *has a unique optimal location decision* $\mathbf{X}^\star$ *under the periodic budgets* $\boldsymbol{\tau}$. *If* $\mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X}^\star) = \{t^\star\}$, *then*

$$\nabla_{\tau_{t^\star}} \theta^\star = \frac{1}{\sum_{s\in[t^\star]} \|[\mathbf{B}_{t^\star}^s]^\top \boldsymbol{f}_{t^\star}(\mathbf{X}_{t^\star}^\star)\|_*};$$

whereas $\nabla_{\tau_t} \theta^\star = 0$ *for all* $t \in [T] \setminus \{t^\star\}$. *If* $|\mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X}^\star)| \ge 2$, *then*

$$\nabla_{\tau_t}^- \theta^\star = \frac{1}{\sum_{s\in[t]} \|[\mathbf{B}_t^s]^\top \boldsymbol{f}_t(\mathbf{X}_t^\star)\|_*} \quad \text{and} \quad \nabla_{\tau_t}^+ \theta^\star = 0 \quad \forall t \in \mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X}^\star);$$

whereas $\nabla_{\tau_t} \theta^\star = 0$ *for all* $t \in [T] \setminus \mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X}^\star)$.

Theorem 3 implies that if there is only a unique period corresponding to the optimal robustness level $\theta^\star$, then $\theta^\star$ is sensitive *only* to both increasing and decreasing ends of the budget for that period. If there are multiple periods corresponding to the optimal robustness level $\theta^\star$, then $\theta^\star$ is sensitive *only* to the decreasing end of the budgets for those periods. Recall that

$$\nabla_\theta Z_t(\mathbf{X}, \theta) = \sum_{s\in[t]} \|[\mathbf{B}_t^s]^\top \boldsymbol{f}_t(\mathbf{X}_t)\|_* \quad \forall t \in [T].$$

Then, the sensitivity of the optimal robustness level with respect to the periodic budgets and that of the worst-case expected cost with respect to the size of the ambiguity set can be connected via

$$[\nabla_\theta Z_1(\mathbf{X}, \theta)), \ldots, \nabla_\theta Z_T(\mathbf{X}, \theta)]^\top \nabla_{\boldsymbol{\tau}}^- \theta^\star = |\mathcal{T}^\star(\boldsymbol{\tau}, \mathbf{X})|.$$

5. Finite-Sample Performance Guarantees

As discussed in Section 3.2, it is more practical to use the historical ERN samples as in Example 4 to construct the (Nested) ambiguity set $\mathcal{F}(\theta)$ because the true distribution of ERN is not easy to specify, whereas a general distribution of ERN may render the problem computationally hard to solve (for instance, in the capacitated case). Accordingly, given a location decision $\mathbf{X}$ solved by our ($\mathcal{BDO}$) model under the (Nested) ambiguity set, it is desirable to have the following *finite-sample performance guarantees* on the confidence of meeting periodic budgets:

$$\bar{\mathbb{P}}^N[\mathbb{E}_{\mathbb{P}}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \mathrm{Q}_t(\mathbf{X}_t, \tilde{\boldsymbol{u}}_t)] \le \tau_t \quad \forall t \in [T]] \ge 1 - \alpha(\theta, N).$$

Here, $\bar{\mathbb{P}}^N$ is the joint distribution of time-series samples $\hat{\boldsymbol{u}}_0, \ldots, \hat{\boldsymbol{u}}_{1-N}, \hat{\boldsymbol{u}}_{-N}, \ldots, \hat{\boldsymbol{u}}_{1-N-P}$ used in the forecast model, $\mathbb{P}$ is the true distribution of the future demand path $(\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T)$, and the level $\alpha(\theta, N)$ approaches zero as N becomes sufficiently large for a given θ.

Finite-sample performance guarantees are typically based on the measure concentration, which is described by the probability that the true distribution falls into the (Nested) ambiguity set, that is, $\bar{\mathbb{P}}^N[\mathbb{P} \in \mathcal{F}(\theta)]$. Because the reference distributions in the Wasserstein balls $\mathcal{B}_t(\theta), t \in [T]$ are the certain same $\hat{\mathbb{Q}}$ constructed from the historical ERNs, it can be shown that $\bar{\mathbb{P}}^N[\mathbb{P} \in \mathcal{F}(\theta)]$ is equal to the probability that the Wasserstein ball $\mathcal{B}_1(\theta)$ contains the true distributions $\mathbb{Q}_t, t \in [T]$ of ERNs $\tilde{\boldsymbol{\xi}}_t$'s.

Lemma 1. *Given* $\mathcal{F}(\theta)$ *in* (Nested) *with* $\hat{\mathbb{Q}}_t = \hat{\mathbb{Q}} \ \ \forall t \in [T]$, *it holds that*

$$\bar{\mathbb{P}}^N[\mathbb{P} \in \mathcal{F}(\theta)] = \bar{\mathbb{P}}^N[d(\mathbb{Q}_t, \hat{\mathbb{Q}}) \le \theta \ \ \forall t \in [T]].$$

With Lemma 1, it is then clear that the key is to investigate the relation between the sample size N and the Wasserstein distances $d(\mathbb{Q}_t, \hat{\mathbb{Q}})$ for $t \in [T]$. We would like to remark that the historical ERN samples of $\hat{\mathbb{Q}}$ (which are determined by the forecast model used) are not necessarily i.i.d. samples from the true distribution $\mathbb{Q}_t$ of $\tilde{\boldsymbol{\xi}}_t$ (to be explained soon). This fact violates the assumptions of the existing measure concentration for Wasserstein balls (Fournier and Guillin 2015). Hence, to derive finite-sample performance guarantees in our context of time series, we need to develop a new theory of measure concentration for our (Nested) ambiguity set by identifying suitable statistical regularity conditions in the time-series setting. In the remainder of this section, we focus on the typical VAR(p) processes and refer to Online Appendix A.2 for the more general VARMA(p, q) process.

Because any K-dimensional VAR(p) can be represented as a Kp-dimensional VAR(1) (see detailed derivation in Online Appendix A.1), VAR(p) shares similar technical difficulties with VAR(1). Hence, we focus on the VAR(1) demand process as follows: for each $t \in [T]$,

$$\tilde{\boldsymbol{u}}_t = \boldsymbol{\kappa}^\dagger + \mathbf{M}_1^\dagger \tilde{\boldsymbol{u}}_{t-1} + \tilde{\boldsymbol{\varepsilon}}_t, \tag{7}$$

where $\boldsymbol{\kappa}^\dagger$ and $\mathbf{M}_1^\dagger$ are the true parameters, and $\tilde{\boldsymbol{\varepsilon}}_1, \ldots, \tilde{\boldsymbol{\varepsilon}}_T$ are i.i.d. random noises governed by the true distribution $\mathbb{Q}^\dagger$. We consider the multivariate least squares estimation and the L_ℓ-norm used in (Nested). Our idea of developing finite-sample performance guarantees for the (Nested) ambiguity set can be outlined as follows. We first point out that the true empirical distribution of ERNs is not observable (Step 1). We then consider instead the non-i.i.d. historical ERNs to construct another empirical distribution as discussed in Example 4, for which the conventional measure concentration (e.g., Fournier and Guillin 2015) cannot be applied directly (Step 2).

Step 1. Based on the multivariate least squares estimation, the estimated process is $\tilde{\boldsymbol{u}}_t = \boldsymbol{\kappa} + \mathbf{M}_1 \tilde{\boldsymbol{u}}_{t-1} + \tilde{\boldsymbol{\xi}}_t$, where the estimator is determined by $(\boldsymbol{\kappa}, \mathbf{M}_1) = \mathbf{U}\mathbf{Z}^\top(\mathbf{Z}\mathbf{Z}^\top)^{-1}$ with $\mathbf{U} = (\hat{\boldsymbol{u}}_0, \hat{\boldsymbol{u}}_{-1}, \ldots, \hat{\boldsymbol{u}}_{1-N})$ and $\mathbf{Z} = (\mathbf{Z}_{-1}, \ldots, \mathbf{Z}_{-N})$ such that $\mathbf{Z}_{-n} = (1, \hat{\boldsymbol{u}}_{-n}^\top)^\top \ \ \forall n \in [N]$. At each period $t \in [T]$ after $\tilde{\boldsymbol{u}}_{t-1}$ materializes to be $\boldsymbol{u}_{t-1}$, plugging the True Demand Process (7) into the estimated counterpart, $\tilde{\boldsymbol{\xi}}_t$ can be represented as

$$\tilde{\boldsymbol{\xi}}_t = \tilde{\boldsymbol{u}}_t - \boldsymbol{\kappa} - \mathbf{M}_1 \boldsymbol{u}_{t-1} = \tilde{\boldsymbol{\varepsilon}}_t + \boldsymbol{\kappa}^\dagger - \boldsymbol{\kappa} + (\mathbf{M}_1^\dagger - \mathbf{M}_1)\boldsymbol{u}_{t-1} \quad \forall t \in [T]. \tag{8}$$

Note that $\tilde{\boldsymbol{\varepsilon}}_1, \ldots, \tilde{\boldsymbol{\varepsilon}}_T$ are i.i.d. random noises governed by the true distribution $\mathbb{Q}^\dagger$. Thus, for every $t \in [T]$, given the estimators $\boldsymbol{\kappa}$ and $\mathbf{M}_1$ as well as the realization $\boldsymbol{u}_{t-1}$, the true empirical distribution of $\tilde{\boldsymbol{\xi}}_t$ should be constructed from i.i.d. samples in the form

$$\hat{\boldsymbol{\varepsilon}}_{1-n} + \boldsymbol{\kappa}^\dagger - \boldsymbol{\kappa} + (\mathbf{M}_1^\dagger - \mathbf{M}_1)\boldsymbol{u}_{t-1} \quad \forall n \in [N],$$

where the historical noises are calculated from the historical demands via $\hat{\boldsymbol{\varepsilon}}_{1-n} = \hat{\boldsymbol{u}}_{1-n} - \boldsymbol{\kappa}^\dagger - \mathbf{M}_1^\dagger \hat{\boldsymbol{u}}_{-n}$. However, because the true parameters ($\boldsymbol{\kappa}^\dagger$ and $\mathbf{M}_1^\dagger$) are not observable, neither is $\hat{\boldsymbol{\varepsilon}}_{1-n}$. Therefore, the true empirical distribution of $\tilde{\boldsymbol{\xi}}_t$ is not observable.

Step 2. Instead, we use the historical ERNs given by

$$\begin{aligned}\hat{\boldsymbol{\xi}}_{1-n} &= \hat{\boldsymbol{u}}_{1-n} - \boldsymbol{\kappa} - \mathbf{M}_1 \hat{\boldsymbol{u}}_{-n} \\ &= \hat{\boldsymbol{\varepsilon}}_{1-n} + \boldsymbol{\kappa}^\dagger - \boldsymbol{\kappa} + (\mathbf{M}_1^\dagger - \mathbf{M}_1)\hat{\boldsymbol{u}}_{-n} \quad \forall n \in [N]\end{aligned} \tag{9}$$

to construct another empirical distribution $\hat{\mathbb{Q}} = \frac{1}{N}\sum_{n\in[N]} \delta_{\hat{\boldsymbol{\xi}}_{1-n}}$. Comparing (8) and (9), it is not hard to see that for every $t \in [T]$ and $n \in [N]$, the historical ERN sample $\hat{\boldsymbol{\xi}}_{1-n}$ follows a distribution that is different from the one that $\tilde{\boldsymbol{\xi}}_t \sim \mathbb{Q}_t$ follows. This is because that $\boldsymbol{u}_{t-1}$ is different from $\hat{\boldsymbol{u}}_{-n}$, albeit each $\hat{\boldsymbol{\varepsilon}}_{1-n}$ is identically distributed as $\tilde{\boldsymbol{\varepsilon}}_t$. As a consequence, the conventional measure concentration, which typically requires the empirical distribution to be constructed from i.i.d. samples governed by the same true distribution, cannot be directly applied to address our interested measure concentration

$$\bar{\mathbb{P}}^N[d(\mathbb{Q}_t, \hat{\mathbb{Q}}) \le \theta \ \ \forall t \in [T]]$$

as stated in Lemma 1. To establish our interested measure concentration, we introduce two intermediate distributions—the true distribution of random noises $\mathbb{Q}^\dagger$ and the empirical distribution of random noises $\hat{\mathbb{Q}}^\dagger = \frac{1}{N}\sum_{n\in[N]} \delta_{\hat{\boldsymbol{\varepsilon}}_{1-n}}$—to bridge $\mathbb{Q}_t$ and $\hat{\mathbb{Q}}$ via the following triangle inequality:

$$d(\mathbb{Q}_t, \hat{\mathbb{Q}}) \le d(\mathbb{Q}_t, \mathbb{Q}^\dagger) + d(\mathbb{Q}^\dagger, \hat{\mathbb{Q}}^\dagger) + d(\hat{\mathbb{Q}}^\dagger, \hat{\mathbb{Q}}).$$

We then investigate for every $t \in [T]$ the Wasserstein distances $d(\mathbb{Q}^\dagger, \hat{\mathbb{Q}}^\dagger)$, $d(\hat{\mathbb{Q}}^\dagger, \hat{\mathbb{Q}})$, and $d(\mathbb{Q}_t, \mathbb{Q}^\dagger)$, respectively. Our development also leverages some statistical regularity conditions on the demand process, the random noises, and the estimated process, which are stated as follows. Online Appendix A provides a topology of the proof sketch (Figure 7) and some technical lemmas.

Assumption 1 (Stability). *The VAR(1) Demand Process (7) is stable. That is, all eigenvalues of $\mathbf{M}_1^\dagger$ have a modulus less than one (Lütkepohl 2005, p. 15). The i.i.d. random noises $\tilde{\boldsymbol{\varepsilon}}_1, \ldots, \tilde{\boldsymbol{\varepsilon}}_t$ are governed by the distribution $\mathbb{Q}^\dagger$ with a finite covariance matrix $\Sigma^\dagger = \mathbb{E}_{\mathbb{Q}^\dagger}[\tilde{\boldsymbol{\varepsilon}}_t \tilde{\boldsymbol{\varepsilon}}_t^\top]$.*

For every $s \ge 0$, let $[\mathbf{M}_1^\dagger]^s = \underbrace{\mathbf{M}_1^\dagger \cdots \mathbf{M}_1^\dagger}_{s}$. The stability condition ensures that $\{[\mathbf{M}_1^\dagger]^s\}_{s\ge 0}$ is an absolutely summable sequence (Lütkepohl 2005, p. 657). That is, there exists a constant h such that

$$h = \sum_{s=0}^{\infty} \|[\mathbf{M}_1^\dagger]^s\|_2 < \infty, \tag{10}$$

where $\|[\mathbf{M}_1^\dagger]^s\|_2 = \sqrt{\sum_{i,j\in[K]} m_{ij}^2}$, with m_{ij} being the (i, j)-entry in the matrix $[\mathbf{M}_1^\dagger]^s$.

Assumption 2 (Light Tail). *There is a constant $a > 1$ such that $\delta = \mathbb{E}_{\mathbb{Q}^\dagger}[\exp(\|\tilde{\boldsymbol{\varepsilon}}_t\|_2^a)] < \infty$.*

Together with the stability condition, the light tail assumption ensures that the demand distribution decays rapidly at the tail, which is critical for our result of measure concentration. The light tail assumption is also commonly required by the conventional measure concentration (see, e.g., Proposition 3). Finally, we make the following assumption to regularize the estimated process.

Assumption 3 (Regularity). *For any sample size $N > K + 1$, there exists a $(K + 1) \times (K + 1)$ matrix*

$$\mathbf{H}_N = \mathbb{E}_{\bar{\mathbb{P}}^N}\left[\left(\frac{\mathbf{Z}\mathbf{Z}^\top}{N}\right)^{-1}\right]$$

such that $\mathbf{Z} = (\mathbf{Z}_{-1}, \ldots, \mathbf{Z}_{-N})$ with $\mathbf{Z}_{-n} = (1, \hat{\boldsymbol{u}}_{-n}^\top)^\top \ \ \forall n \in [N]$.

In particular, as sample size N approaches infinity, the matrix $\lim_{N\to\infty}\mathbf{H}_N$ also exists.

Assumption 3 ensures the asymptotic consistency of the multivariate least squares estimators. In fact, under some additional mild conditions, Assumption 3 follows from Assumption 1; see theorem 8.2.3 in Fuller (1976) for more details.

Now, we are ready to investigate the Wasserstein distances $d(\mathbb{Q}^\dagger,\hat{\mathbb{Q}}^\dagger)$, $d(\hat{\mathbb{Q}}^\dagger,\hat{\mathbb{Q}})$, and $d(\mathbb{Q}_t,\mathbb{Q}^\dagger)$. To begin with, we look at $d(\mathbb{Q}^\dagger,\hat{\mathbb{Q}}^\dagger)$—the Wasserstein distance between the true distribution and the empirical distribution of the random noises. Because $\hat{\mathbb{Q}}^\dagger$ is constructed from i.i.d. samples $\hat{\boldsymbol{\varepsilon}}_{1-n}$, $n\in[N]$, we directly have the following result from Fournier and Guillin (2015) without proof.

Proposition 3 (Fournier and Guillin 2015, Theorem 2). *Under Assumption 2, for any $\theta>0$ and any $K>2$, we have*

$$\bar{\mathbb{P}}^N[d(\mathbb{Q}^\dagger,\hat{\mathbb{Q}}^\dagger)>\theta/3]\le\alpha_1(\theta,N)=\begin{cases}c_1\exp(-c_2N\theta^K) & \text{if } 0<\theta\le 3\\ c_1\exp(-c_2N\theta^a) & \text{if } \theta>3,\end{cases}\tag{11}$$

where c_1 and c_2 are positive constants that only depend on a, δ, and K.

We next study $d(\hat{\mathbb{Q}}^\dagger,\hat{\mathbb{Q}})$—the Wasserstein distance between the empirical distribution of the historical ERNs and the empirical distribution of the random noises.

Proposition 4. *Under Assumptions 1, 2, and 3, for any $\theta>0$, we have*

$$\bar{\mathbb{P}}^N[d(\hat{\mathbb{Q}}^\dagger,\hat{\mathbb{Q}})>\theta/3]\le\alpha_2(\theta,N)+\frac{\alpha_3}{N},$$

where

$$\alpha_2(\theta,N)=\left[\frac{12\max\{K^2,K^{1/\ell+3/2}\}}{\theta}\right]\cdot\sqrt{\frac{\mathrm{tr}(\Sigma^\dagger)\mathrm{tr}(\mathbf{H}_N)}{N^{(3a-2)/(3a)}}}\tag{12}$$

(which is related to the L_ℓ-norm and satisfies $\alpha_2(\theta,N)\to 0$ as $N\to\infty$) and

$$\alpha_3=\|(\mathbf{I}_K-\mathbf{M}_1^\dagger)^{-1}\boldsymbol{\kappa}^\dagger\|_2^{6a}+6!(\delta-1)h^{6a}\tag{13}$$

(which is a positive constant defined by the parameters a and δ of the light-tailed distribution and the constant h defined in (10)).

Finally, we investigate $d(\mathbb{Q}_t,\mathbb{Q}^\dagger)$—the Wasserstein distance between the true distribution of ERNs and that of the random noises.

Proposition 5. *Under Assumptions 1, 2, and 3, for any $\theta>0$, we have*

$$\bar{\mathbb{P}}^N[d(\mathbb{Q}_t,\mathbb{Q}^\dagger)>\theta/3]\le\alpha_2(\theta,N)+\frac{T\alpha_3}{N^2}\quad\forall t\in[T].$$

We emphasize that the technical treatments for establishing the measure concentration in Propositions 4 and 5 are different from those of the extant studies by Qi et al. (2022a) and Kannan et al. (2023). In particular, we consider the multivariate time-series uncertainty with an underlying stochastic process, whereas the studies by Qi et al. (2022a) and Kannan et al. (2023) focus on a cross-sectional uncertainty with an underlying true value. In our setting, the historical demand samples are not only used to estimate the time-series model but are also treated as explanatory variables to forecast future demand (i.e., explained variables). In other words, the residuals' distribution depends on both the model estimators and the distribution of previous demands. Therefore, their measure concentration results for cross-sectional uncertainty cannot be directly applied to our setting of time-series uncertainty, and to establish our measure concentration, we leverage the stability of the underlying stochastic process, the regularity of the estimated stochastic process, and the estimation method of the stochastic process. Besides, some popular time-series models (e.g., VARMA) involve partial explanatory variables as nonobservable noises, whose coefficients therefore require even more complicated estimation methods, raising challenges to analyze the measure concentration based on the Wasserstein distance. We will elaborate on this with more details in Online Appendix A.2.

Equipped with Propositions 3–5, we are now ready to establish the measure concentration for our (Nested) ambiguity set at a given radius level θ, which further establishes the desired finite-sample performance guarantees on the confidence of meeting budgets. Here, we also assume that the number N of historical demands is sufficiently larger than the length T of the planning horizon to avoid the measure concentration being trivial.

Theorem 4. *Under Assumptions 1, 2, and 3, for any $\theta>0$, we have*

$$\bar{\mathbb{P}}^N[\mathbb{P}\in\mathcal{F}(\theta)]\ge 1-T\cdot\left(\alpha_1(\theta,N)+2\alpha_2(\theta,N)+\frac{\alpha_3}{N}+\frac{T\alpha_3}{N^2}\right).$$

Theorem 5 (Finite-Sample Performance Guarantees). *Under Assumptions 1, 2, and 3, for any feasible $(\mathbf{X},\theta)$ in ($\mathcal{BDO}$), the confidence of meeting the budget satisfies*

$$\begin{aligned}&\bar{\mathbb{P}}^N[\mathbb{E}_{\mathbb{P}}[\boldsymbol{c}_t^\top\boldsymbol{x}_t+\mathrm{Q}_t(\mathbf{X}_t,\tilde{\boldsymbol{u}}_t)]\\&\le\tau_t\quad\forall t\in[T]]\\&\ge 1-T\cdot\left(\alpha_1(\theta,N)+2\alpha_2(\theta,N)+\frac{\alpha_3}{N}+\frac{T\alpha_3}{N^2}\right).\end{aligned}$$

Here, $\alpha_1(\theta,N)$, $\alpha_2(\theta,N)$ and α_3 are defined in (11), (12), and (13), respectively.

Given a radius level θ, the confidence approaches 1 as $N\to\infty$. This establishes the statistical validity of our (Nested) ambiguity set. In addition, confidence increases in θ because both $\alpha_2(\theta,N)$ and $\alpha_3(\theta,N)$ are

decreasing in θ. This means that given a finite number of samples, higher confidence in meeting budgets corresponds to a larger radius θ—the key parameter that our ($\mathcal{BDO}$) model strives to maximize.

To conclude this section, we note that finite-sample performance guarantees can be readily extended to the VAR(p) process, as it can be equivalently viewed as a multidimensional VAR(1) process. Indeed, for a stable VAR(p) process and the multivariate least squares estimation as in Example 1, under the corresponding assumptions on the transformed VAR(1) process, its random noises, and the estimated process, all preliminary lemmas and propositions in the proof sketch (Figure 7 of Online Appendix A.1) can be derived without additional technical difficulties. In Online Appendix A.2, we further extend finite-sample performance guarantees of ($\mathcal{BDO}$) to the VARMA(p, q) process. The statistical regularity conditions on the process stability (Assumption 1/Assumption 4), the light-tailed random noise (Assumption 2), and the estimated process (Assumption 3/Assumption 5) are consistent for both VAR(p) and VARMA(p, q) processes. These conditions are self-contained for developing finite-sample performance guarantees herein, but not necessary for results on the operational insights, tractable reformulations, and solution schemes.

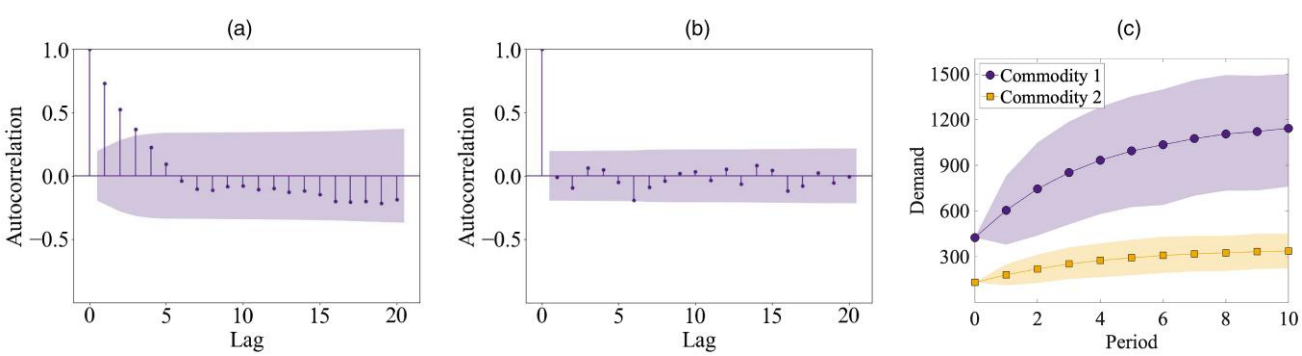
6. Numerical Experiments

Instances for our numerical experiments are generated from the Civil Aeronautics Board (CAB) data set originated from O'Kelly (1987).[6] The data set contains a collection of cities in the United States with Euclidean distances and directed demands between each pair of cities. In particular, we randomly select $K = 100$ pairwise directed demands between the first 50 cities as commodities, and we take the next I cities as the potential hubs to be selected with $I \in \{10, 15, 20\}$. We consider the planning horizon $T \in \{5, 10, 15\}$. All optimization problems are solved with Gurobi 10.0.1 on an Intel Core™ 3.60-GHz processor with 64 GB memory. The data and source code for reproducing the results can be found in the Online Appendix.

We set the initial demand $\hat{\boldsymbol{u}}_0$ to the value in the CAB data set, and we generate historical demands $\hat{\boldsymbol{u}}_{-1}, \ldots, \hat{\boldsymbol{u}}_{1-N}$ and a presample $\hat{\boldsymbol{u}}_{-N}$ from the true VAR(1) process specified in (7), where $\boldsymbol{\kappa}^\dagger$ and $\mathbf{M}_1^\dagger$ are randomly generated intercept and coefficient matrix, respectively, and for every $t = -1, \ldots, -N$, the random noise $\tilde{\boldsymbol{\varepsilon}}_t$ follows the uniform distribution in $[-0.6\hat{\boldsymbol{u}}_0, 0.6\hat{\boldsymbol{u}}_0]$. Based on these historical demands, we apply techniques in time-series analysis to obtain the (Time Series) model. Observe from Figure 1 that the temporal dependence of the first commodity demand can be decomposed by independent ERNs over periods. In addition, based on the estimated model, we can forecast the future demand path. The future demand fluctuations of the first two commodities increase along the future 10 periods, showing the propagation of uncertainty over periods. In the subsequent experiments, we incorporate the estimated time-series model into the (Nested) ambiguity set.

Our experiments consist of four parts: (i) analyzing the impacts of periodic budgets on the optimal location decision (Section 6.1), (ii) showing the advantage of incorporating time series in the ambiguity set (Section 6.2), (iii) demonstrating the finite-sample performance (Section 6.3), and (iv) comparing solution performance associated with VAR and VARMA models (Section 6.4). To further validate the effectiveness of our ($\mathcal{BDO}$) model with real-world time-series data, we conduct another numerical experiment using the monthly demand data set of the U.S. domestic flights among 563 cities from 1990 to 2009, where the profile of the optimal location decision justifies the performance of the model, which exhibits similar patterns as that under the time-series demands generated using the CAB data set in Online Appendix D.3; see Appendix Online D.5 for more details.

Figure 1. (Color online) Forecasting Results of the Time-Series Model



Notes. (a) Autocorrelation function of the historical demands of the first commodity. (b) Autocorrelation function of the corresponding ERNs. (c) Forecast obtained from the estimated time-series model for the first two commodities, where the shaded areas represent the 5%–95% confidence interval.

6.1. Location Decision Under Different Budget Settings

We use $N = 10^3$ historical demands for training and set the planning horizon to $T = 10$. We study the performance of the ($\mathcal{BDO}$) model over the (Nested) ambiguity set under different budget settings wherein periodic budgets increase at a growth rate r within the planning horizon, where $r \in \{-2\%, 0\%, 2\%\}$. That is, the budget for each period follows

$$\tau_t = (1+r)^{t-1}\tau_1 \quad \forall t \in [T].$$

Specifically, we set the total budget of ($\mathcal{BDO}$) to $\sum_{t\in[T]}\tau_t = \rho\tau_0$ with a normalized budget level $\rho \geq 1$, and the optimal expected cost τ_0 under the empirical distribution—as a baseline—given by

$$\tau_0 = \min_{\mathbf{X}\in\mathcal{X}} \mathbb{E}_{\hat{\mathbb{Q}}_{[T]}}\left[\sum_{t\in[T]} \boldsymbol{c}_t^\top \boldsymbol{x}_t + \tilde{\boldsymbol{u}}_t^\top \boldsymbol{f}_t(\mathbf{X}_t)\right].$$

We compare against the ($\mathcal{DRO}$) model that seeks the optimal location decision with the least worst-case expected total cost over the same (Nested) ambiguity set (see Online Appendix C.1 for the detailed reformulation and derivation). For a fair comparison, we adopt the fivefold cross-validation to tune the parameter of both models, that is, ρ in ($\mathcal{BDO}$) and θ in ($\mathcal{DRO}$). Specifically, we select ρ from $\{1, 1.02, 1.04, 1.06, 1.08, 1.1, 1.12, 1.14, 1.16, 1.18, 1.2\}$ and θ from $\{0, 20, 40, 60, 80, 100, 200, 400, 600, 800, 1,000\}$. We note that the optimal solution of ($\mathcal{DRO}$) is insensitive to the radius θ when it is small, and the best parameter of ($\mathcal{DRO}$) is $\theta = 0$, that is, no robustness. In addition, we also investigate our ($\mathcal{BDO}$) model under some not well-planned periodic budgets (e.g., smaller and larger magnitude ρ of periodic budgets and more significant fluctuation r of periodic budgets). The results are relegated to Online Appendix D.4.

As shown in Table 1, the optimal solutions in both models prefer to construct more hubs in the first period because they can serve longer. Compared with the ($\mathcal{DRO}$) solution that sets up all selected hubs in the first period, the ($\mathcal{BDO}$) solution in each case of growth rate r tends to postpone the setup plan evenly to the subsequent periods so as to mitigate the risk of heavy upfront investment in the first period.

With the optimal solutions obtained, we analyze the out-of-sample performance. From the true demand process, we generate 10^4 testing samples of the demand path $(\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T)$. We compare the ($\mathcal{BDO}$) and ($\mathcal{DRO}$) models in terms of the periodic costs and the total cost. We first look at the periodic costs. Figure 2 shows that the first-period cost of the ($\mathcal{BDO}$) model is significantly less than that of ($\mathcal{DRO}$), and the former yields periodic costs that are more stable than the latter. To measure the fluctuation of periodic costs more precisely, we define a criterion—normalized range of periodic costs—as follows:

$$\frac{\texttt{maximal cost over periods} - \texttt{minimal cost over periods}}{\texttt{minimal cost over periods}}.$$

The results are summarized in the boxplots of Figure 2, showing that under every budget setting we have considered, the fluctuation of periodic costs with the ($\mathcal{BDO}$) model is less than that with the ($\mathcal{DRO}$) model. This justifies that ($\mathcal{BDO}$) can yield an optimal location decision leading to a more stable cost flow over the planning horizon. Turning to the total cost, Figure 3 shows that our ($\mathcal{BDO}$) model may pay a higher out-of-sample total cost than that of $\mathcal{DRO}$ (that is, on average, 2%–2.74% more depending on the growth rate). This cautions an extra price of controlling the fluctuation in cost flow as the trade-off.

Table 1. Profiles of the Optimal Location Decision

		Selected hub(s) in each period					
Model	r	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$–10	Total
$\mathcal{BDO}$	−2%	{2,3}	{1}	{15}	{12}	—	5
	0%	{2,3}	{1}	{15}	{12}	—	5
	2%	{2,3}	{4,15}	{12,14}	—	—	6
$\mathcal{DRO}$	—	{2,3,4,12,15}	—	—	—	—	5

6.2. Advantage of Incorporating Time Series

We show the value of incorporating (Time Series) into the ambiguity set as in our ($\mathcal{BDO}$) model by comparing it with a traditional budget-driven optimization model—denoted as ($\mathcal{BDO}_{\mathrm{W}}$)—that considers a traditional Wasserstein ambiguity set with historical samples of demand path. Specifically, we construct a type-1 Wasserstein ball for the random demand $\tilde{\mathbf{U}}_T = (\tilde{\boldsymbol{u}}_1, \ldots, \tilde{\boldsymbol{u}}_T)$:

$$\mathcal{F}_{\mathrm{W}}(\theta) = \{\mathbb{P} \in \mathcal{P}(\mathbb{R}_+^{K\times T}) \mid d(\mathbb{P}, \hat{\mathbb{P}}) \leq \theta\},$$

where the empirical distribution $\hat{\mathbb{P}} = \frac{1}{N}\sum_{n\in[N]}\delta_{\hat{\mathbf{U}}^n}$ is constructed from historical demands

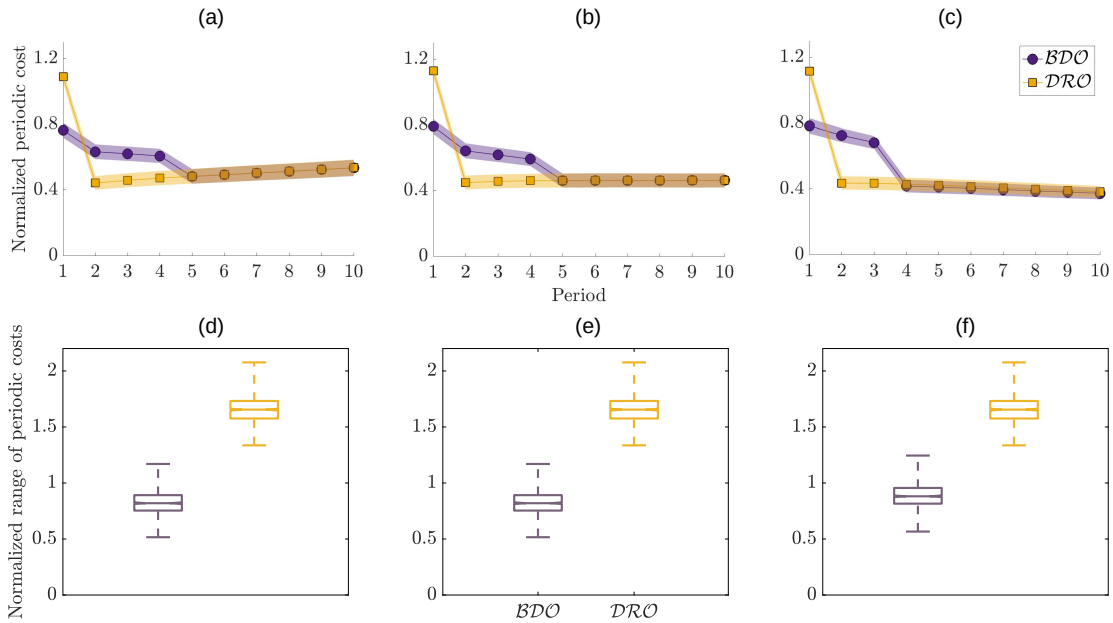
$$\hat{\mathbf{U}}^n = (\hat{\boldsymbol{u}}_0, \ldots, \hat{\boldsymbol{u}}_0) \;\; \forall n = 1, \ldots, T-1 \quad \text{and}$$
$$\hat{\mathbf{U}}^n = (\hat{\boldsymbol{u}}_{1-n}, \hat{\boldsymbol{u}}_{2-n}, \ldots, \hat{\boldsymbol{u}}_{T-n}) \;\; \forall n = T, \ldots, N.$$

Here, for a fair comparison with the same number N of demand paths, we set the first $T-1$ demand paths to be identical. Equipped with the Wasserstein ambiguity set $\mathcal{F}_{\mathrm{W}}(\theta)$, a budget-driven ($\mathcal{BDO}_{\mathrm{W}}$) model is captured by

$$\begin{aligned} \text{maximize} \quad & \theta \\ \text{subject to} \quad & \sup_{\mathbb{P}\in\mathcal{F}_{\mathrm{W}}(\theta)} \mathbb{E}_{\mathbb{P}}[\boldsymbol{c}_t^\top \boldsymbol{x}_t + \mathrm{Q}_t(\mathbf{X}_t, \tilde{\boldsymbol{u}}_t)] \leq \tau_t \quad \forall t \in [T] \\ & \mathbf{X} \in \mathcal{X}, \theta \geq 0. \end{aligned} \tag{$\mathcal{BDO}_{\mathrm{W}}$}$$

By exploiting the appealing structure of the uncapacitated MHL problem, constraints in the ($\mathcal{BDO}_{\mathrm{W}}$) model also admit tractable reformulations (see details in Online Appendix C.2). The only difference between

Figure 2. (Color online) Fluctuations of the Periodic Costs



Notes. (a)–(c) Magnitude of the normalized periodic costs, and (d)–(f) corresponding boxplots of the normalized range of periodic costs. (a) and (d) $r = -2\%$; (b) and (e) $r = 0\%$; (c) and (f) $r = 2\%$. The normalized cost in each period is the realized cost divided by the periodic budget, and the shaded areas represent the 5%–95% confidence intervals.

($\mathcal{BDO}$) and ($\mathcal{BDO}_{\mathrm{W}}$) is whether to incorporate the time-series model when characterizing uncertain periodic demands.

We focus on the growth rate $r = 0\%$ and consider 10^3 (respectively, 10^4) samples for training (respectively, testing). Figure 4 summarizes the out-of-sample performance of ($\mathcal{BDO}$) and ($\mathcal{BDO}_{\mathrm{W}}$) models, measured by the joint probability of meeting all periodic budgets:

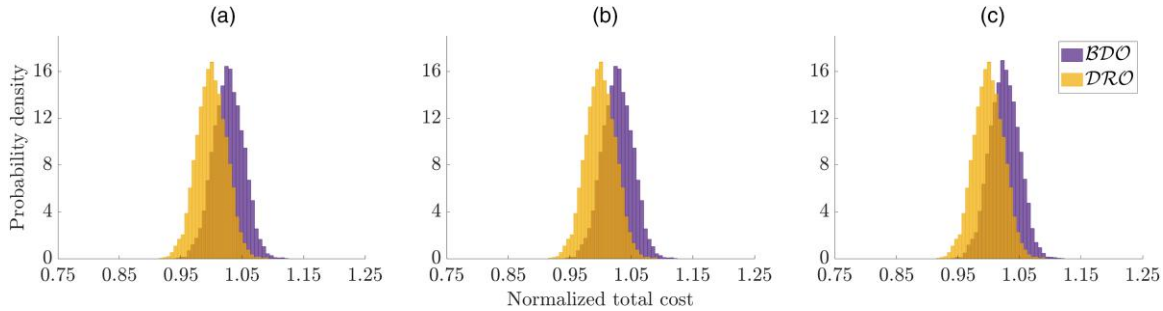
$$\frac{\texttt{number of testing sample paths with periodic costs less than periodic budgets}}{\texttt{number of testing sample paths}}. \tag{14}$$

We observe that under different values of the predetermined normalized budget level ρ, our proposed ($\mathcal{BDO}$) model outperforms that of ($\mathcal{BDO}_{\mathrm{W}}$). For many values of ρ, for example, $\rho \in [1.04, 1.08]$, the constraint system of ($\mathcal{BDO}$) for all cases of $T \in \{5, 10, 15\}$ can guarantee an optimal solution, whereas the constraint system of ($\mathcal{BDO}_{\mathrm{W}}$) is infeasible. In other words, our ($\mathcal{BDO}$) model—incorporating (Time Series)—can yield a feasible solution with smaller worst-case expected periodic costs that meet the corresponding periodic budgets. When the budget level is large (e.g., $\rho \geq 1.12$), for all cases of T, the ($\mathcal{BDO}$)'s probability of meeting budgets is also higher than that of ($\mathcal{BDO}_{\mathrm{W}}$). These observations well justify the value of incorporating (Time Series) in our ($\mathcal{BDO}$) model.

6.3. Finite-Sample Performance

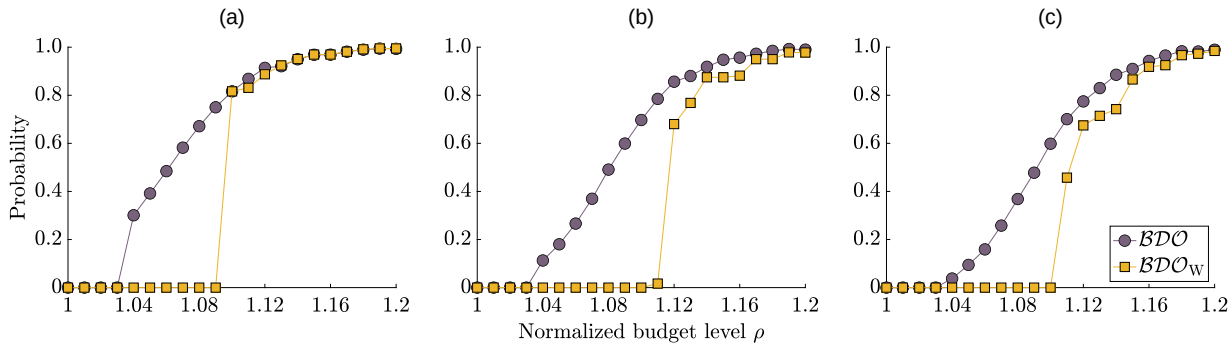
We evaluate the finite-sample performance of our ($\mathcal{BDO}$) model by investigating the performance of the optimal location decision trained by different numbers

Figure 3. (Color online) Probability Density Distribution of the Total Cost



Notes. (a) $r = -2\%$; (b) $r = 0\%$; (c) $r = 2\%$. For each testing sample, the normalized total cost is the realized cost divided by the baseline τ_0.

Figure 4. (Color online) Joint Probability of Meeting All Periodic Budgets



Notes. (a) $T = 5$. (b) $T = 10$. (c) $T = 15$. A zero probability means that the model is infeasible.

of historical demands. We focus on the growth rate $r = 0\%$ and the normalized budget level $\rho = 1.12$, and for each length of planning horizon T, we generate M (which ranges from 50 to 1,000) historical demands to train the model. In particular, for each value of M, we first estimate a VAR(1) time-series model and then solve the ($\mathcal{BDO}$) model equipped with the estimated time series. Based on 10^4 testing samples of demand path, in Figure 5, we plot, for the optimal location decision, its joint probability of meeting all periodic budgets and individual probabilities of meeting each periodic budget.

First, we observe that for all cases of $T \in \{5, 10, 15\}$, the joint probability of meeting all periodic budgets converges well with no more than 40% of the total amount (i.e., 1,000) of historical demands. Given the same sample size M, the joint probability under a longer planning horizon (i.e., $T = 15$) is relatively lower than that under a shorter planning horizon (i.e., $T = 5$). This is expected by noting that under a longer planning horizon, there are more periodic budgets to be met jointly. As for the individual probabilities, we observe that as the sample size M increases, more members of them approach one. These observations reasonably justify the finite-sample performance of our ($\mathcal{BDO}$) model in meeting budgets.

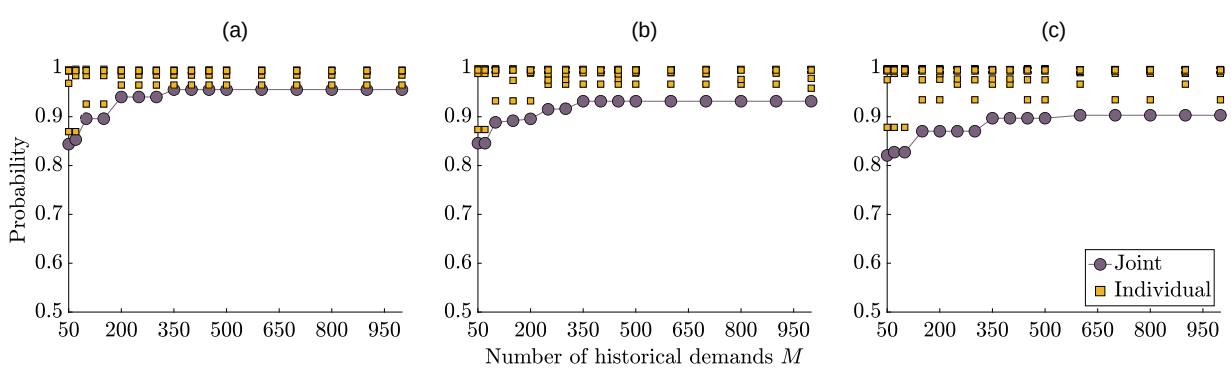
6.4. Comparison of VAR and VARMA

We compare the performance of our ($\mathcal{BDO}$) model equipped with different time-series models: VARMA and VAR. In this section, we generate 10^3 historical demands for training and 10^4 testing samples from an underlying VARMA(1, 1) process:

$$\tilde{\boldsymbol{u}}_t = \boldsymbol{\kappa}^\dagger + \mathbf{M}_1^\dagger \tilde{\boldsymbol{u}}_{t-1} + \mathbf{L}_1^\dagger \tilde{\boldsymbol{\xi}}_{t-1} + \tilde{\boldsymbol{\xi}}_t.$$

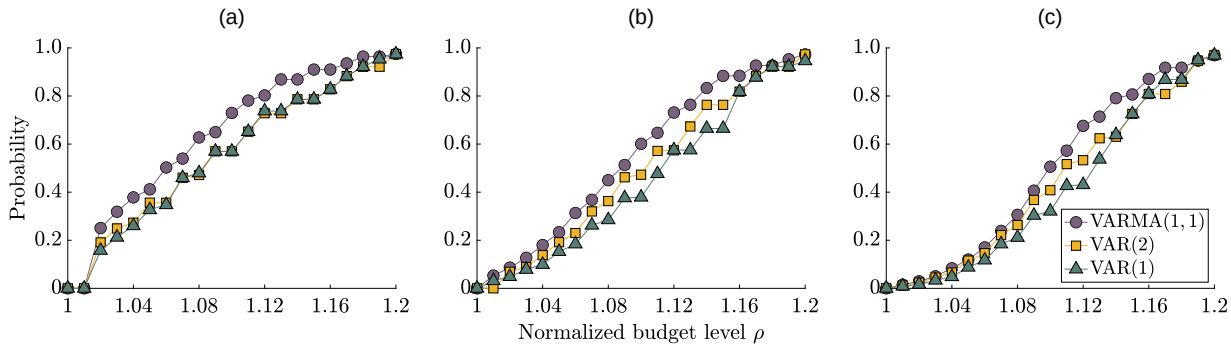
Here, $\boldsymbol{\kappa}^\dagger$, $\mathbf{M}_1^\dagger$, and $\mathbf{L}_1^\dagger$ are randomly generated parameters. Our consideration of choosing VARMA as the underlying process follows from the fact that the VAR model can be treated as a special VARMA model, and if choosing VAR instead, then the estimated VAR and VARMA models would be similar. We apply VARMA(1, 1), VAR(2), and VAR(1) to estimate the time-series models from historical demands, respectively. For each combination of (ρ, T), we solve the ($\mathcal{BDO}$) model with the corresponding estimated time-series model to obtain the optimal location decision. We observe in Figure 6 that for all combinations of ρ and T,

Figure 5. (Color online) Joint Probability of Meeting All Periodic Budgets and Individual Probabilities of Meeting Each Period Budget



Notes. (a) $T = 5$. (b) $T = 10$. (c) $T = 15$.

Figure 6. (Color online) Joint Probability of Meeting All Periodic Budgets



Notes. (a) $T = 5$. (b) $T = 10$. (c) $T = 15$.

($\mathcal{BDO}$) with the VARMA model outperforms the other two that are based on the VAR model. Meanwhile, in most cases, ($\mathcal{BDO}$) with VAR(2) outperforms that with VAR(1). This is intuitive by noting that VAR(2)—more general than VAR(1)—is less misspecified from the underlying model VARMA(1,1). These observations reveal the importance of incorporating the right time-series model in ($\mathcal{BDO}$) and caution that the performance could be depreciated if the time-series model is misspecified.

7. Conclusion

We introduce to the multiperiod hub location problem a novel (Nested) ambiguity set that uses time series to capture the underlying stochastic demand process. Our budget-driven optimization model, ($\mathcal{BDO}$), possesses attractive finite-sample performance guarantees on the confidence of meeting budgets, thus mitigating the risk of cost fluctuations throughout the planning horizon.

Finite-sample performance guarantees of our (Nested) ambiguity set based on the type 1 Wasserstein distance can be readily extended to other types of Wasserstein distance (e.g., type ∞) but remains open to more complex time-series models. It is challenging yet interesting to ascertain its performance with other time-series models (e.g., vector error correction and multivariate generalized autoregressive conditional heteroskedasticity (GARCH)). Another future direction is to design fast algorithms to enhance the scalability of the capacitated model, which does not scale so well with the length of planning horizon T and the sample size N. A promising approach is to explore the convexity of the approximate worst-case expected cost function as well as its decomposability with respect to periods and to incorporate with the Benders decomposition (see, e.g., Wang et al. 2021a and Zheng et al. 2021, which consider Benders decomposition to address location problems).

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Endnotes

[1] IUT-Global, as the largest food waste recycling company in Singapore, survived only three years and ceased to operate in 2011. Because of an overly optimistic projection of the collectible feedstock for waste-to-energy treatment in the subsequent periods, the firm invested in building a phase II waste-to-energy plant in 2009 that doubled the capacity of the phase I plant. However, as reported by The Straits Times (2011), the actual collected feedstock was only 15% of the projected level, and only one-third of the plant's capacity was effectively utilized. Thus, the firm—with a much lower-than-expected revenue flow—could not afford the high upfront cost for facility installation and expansion in a sustainable manner.

[2] To be consistent with our context of cost, we will use "budget" over "target" throughout our paper.

[3] The number of presamples depends on the time-series model used for estimation. For instance, the VAR(p) model needs $P = p$ presamples, whereas the VARMA(p, q) model requires $P = \max\{p, q\}$ presamples.

[4] Our $\mathcal{BDO}$ model that maximizes the robustness in terms of the size of the ambiguity set follows the spirit of the satisficing approach proposed by Brown and Sim (2009). This also shares a similar spirit of robust satisficing recently proposed in Long et al. (2022), which focuses on minimizing the fragility measured by the worst-case magnitude of expected target violation under a distribution normalized by the statistical distance of the nominal distribution. We also emphasize that—as we are to show in Section 4.1—maximizing the size of the ambiguity set in our hub location problem context (uncapacitated case) leads essentially to optimizing a Sharpe ratio–type criterion over the worst case among all periods, which enjoys an insightful interpretation.

[5] Based on these project-level proposals, the final budget could be comprehensively adjusted by incorporating the strategic, financial, and operational information on the stakeholders of the project (Mihm 2010, Zwikael et al. 2018).

[6] See https://www.researchgate.net/publication/269396247_cab100_mok.

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Jie Hu is an assistant professor at the School of Economics and Management, Beijing Jiaotong University. His research interests include data-driven optimization, distributionally robust optimization, facility location, and supply chain management.

Zhi Chen is an assistant professor at the Department of Decisions, Operations, and Technology, CUHK Business School, The Chinese University of Hong Kong. His research focuses on developing models and designing algorithms for analytics under uncertainty with different levels of data availability as well as applications in business, economics, finance, and operations.

Shuming Wang is a professor at the School of Economics and Management, University of Chinese Academy of Sciences (UCAS). His research interests include robust optimization, stochastic programming, and data-driven analytics, with applications in location science and transportation, supply chain management, financial risk management, and healthcare operations.